# THE INCOMPLETENESS OF OBSERVATION

## Why Quantum Mechanics and General Relativity Cannot Be Unified From Within

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**Abstract**

Physical observers are strictly subsystems of the deterministic universes they measure, permanently barred from accessing degrees of freedom beyond their causal reach. This paper proves that the structural constraints imposed by this epistemic reduction necessarily give rise to quantum mechanics. We show that an observer coupled to a slow, high-capacity hidden sector on a finite-dimensional configuration space must describe the visible sector using P-indivisible stochastic dynamics. By the stochastic-quantum correspondence, this reduction is mathematically equivalent to unitarily evolving quantum mechanics.

Crucially, the converse also holds: any quantum system, realized as a deterministic dilation, requires precisely these conditions — non-trivial coupling, slow-bath memory, and sufficient hidden-sector capacity — establishing a full equivalence between quantum mechanics and embedded observation. Consequently, the Schrödinger equation, the Born rule, and Bell inequality violations emerge as structural consequences of this equivalence, requiring no independent quantum postulates.

Applying this theorem to the cosmological horizon, which provides an exact physical realization of these causal conditions, uniquely determines the emergent action scale. Partition-relativity, dimensional analysis, and the thermodynamic consistency of the horizon fix $\hbar = c^3\epsilon^2/(4G)$, establishing the geometric discreteness scale at $\epsilon = 2\,l_p$ and recovering the exact Bekenstein–Hawking entropy formula. Furthermore, this framework intrinsically dissolves the cosmological constant problem: the $10^{122}$ discrepancy is precisely $S_{\mathrm{dS}}$, representing the information compression ratio of the emergent quantum description rather than a catastrophic fine-tuning failure. Falsifiable predictions include dark energy evolution in Running Vacuum

Model form ($\Lambda_{\text{eff}} = \Lambda_0 + \nu H^2$) and distinct gravitational wave echoes near black hole horizons.

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# PART I: THE GENERAL THEOREM

## 1 INTRODUCTION

### 1.1 The Problem of Embedded Observation

Physical observers are subsystems of the systems they measure. This elementary fact has consequences that have not been traced. An observer embedded in a deterministic universe cannot access the complete state: degrees of freedom beyond the observer's causal reach are permanently hidden, and the observer's description of the visible sector is necessarily a reduced one obtained by marginalizing over the hidden sector. The question is what structural constraints that reduction imposes on the form of the observer's physical laws.

The observer's epistemic situation has been foregrounded before. QBism treats quantum states as expressions of an agent's beliefs; relational quantum mechanics defines states relative to interacting subsystems; 't Hooft's cellular automaton interpretation [1] derives quantum behavior from deterministic dynamics through information loss. These programs take observer-dependence as an interpretive starting point or, in 't Hooft's case, derive it from a specific microphysical model. The present framework differs in that it identifies *necessary and sufficient conditions* under which any embedded observer in any deterministic system necessarily sees quantum mechanics, independent of the system's specific physical content — and proves that these are the *only* conditions under which quantum mechanics arises from a deterministic embedding.

### 1.2 Axioms

The framework rests on three axioms. None reference quantum mechanics, general relativity, or any specific physical theory.

1. **Deterministic dynamics.** The total system evolves deterministically on a phase space $\Gamma$. The phase-space density $\rho(q, p, t)$ evolves via the Liouville equation:

$$\frac{\partial \rho}{\partial t} = \{H_{\text{tot}}, \rho\} \equiv \mathcal{L}\rho \tag{1}$$

where $\mathcal{L}$ is the Liouville operator and $\{\cdot, \cdot\}$ denotes the Poisson bracket.

**Finiteness condition.** The effective configuration space of the visible sector is finite-dimensional. A discreteness scale $\epsilon$ provides a finite minimal cell volume, ensuring that any finite spatial region contains at most a finite number of independent cells. In Part I, $\epsilon$ is a free parameter; its value is determined *a posteriori* by self-consistency (§6), which pins $\epsilon = 2\,l_p$ exactly. In Part II, the finiteness condition is not an additional assumption but a consequence of the partition geometry: the cosmological horizon has finite area $A = 4\pi c^2/H^2$ (from GR alone), and the number of modes that couple through a finite-area boundary is $A/\epsilon^2$, which is finite for any $\epsilon > 0$. The stochastic process that the trace-out generates therefore lives on a finite-dimensional space regardless of whether the total phase space extends infinitely beyond the horizon. This condition enters the proof at a single point: the stochastic-quantum correspondence (§3.1), which requires a finite configuration space. Independent support comes from holographic arguments that the Hilbert space of quantum gravity is locally finite-dimensional [27], establishing that this condition aligns naturally with known holographic bounds rather than serving as an ad hoc lattice assumption.

2. **Causal partition.** $\Gamma$ is partitioned into a visible sector $\Gamma_V$ accessible to the observer and a hidden sector $\Gamma_H$ that is not:

$$\Gamma = \Gamma_V \times \Gamma_H \tag{2}$$

The partition is fixed on timescales relevant to the observer (it may drift on much longer timescales, but such drift is not required for the theorem). The total Hamiltonian decomposes as:

$$H_{\text{tot}} = H_V + H_H + H_{\text{int}} \tag{3}$$

The product decomposition $\Gamma_V \times \Gamma_H$ is a simplification. A real observer's epistemic access is graded: some degrees of freedom are measured precisely, others poorly, and others not at all. The axiom requires that this graded access be well-approximated by a sharp partition — that there exist a clean division into accessible and inaccessible sectors such that cross-partition correlations enter only through $H_{\text{int}}$. This is a condition on the physics, not just the observer: it is satisfied when a physical boundary (causal, thermodynamic, or spatial) enforces a separation sharp enough that the residual leakage across the boundary can be absorbed into the interaction term. The theorem applies to any system where such an approximation holds; Part II identifies a physical realization where the sharpness is exact.

3. **Classical probability.** The observer's predictions are classical expectation values over the phase space; standard results of probability theory (the law of large numbers, the central limit theorem) are the primary statistical tools.

These axioms contain no quantum postulates — no Hilbert spaces, no commutation relations, no Born rule. The framework's central claim is that quantum mechanics *emerges* from these premises under the conditions specified below (§1.3).

## 1.3 Conditions on the Hidden Sector

The theorem requires three conditions on the relationship between the visible and hidden sectors. These are stated as sufficient conditions for the emergence of quantum mechanics; §3.3 proves they are also necessary, establishing a full equivalence.

**(C1) Non-zero coupling.** The interaction Hamiltonian satisfies $H_{\text{int}} \neq 0$. The visible and hidden sectors are dynamically coupled, not merely co-existing. The coupling is bidirectional: visible-sector transitions influence the hidden-sector state, and the hidden-sector state influences visible-sector transition probabilities.

**(C2) Slow-bath timescale separation.** The hidden sector's correlation time $\tau_B$ vastly exceeds the visible sector's dynamical timescale $\tau_S$:

$$\tau_S \ll \tau_B \tag{4}$$

This is the *inverse* of the standard Markovian regime, which requires $\tau_B \ll \tau_S$. The hidden sector evolves on timescales far exceeding those of any process accessible to the observer.

**(C3) Sufficient capacity.** The hidden sector has enough degrees of freedom that visible-sector interactions do not appreciably perturb its state on timescales $\ll \tau_B$. Precisely: if $N_H$ and $N_V$ denote the number of independent degrees of freedom in the hidden and visible sectors respectively, then $N_H/N_V$ is large enough that the cumulative imprint of all visible-sector transitions during any observationally relevant interval occupies a negligible fraction of the hidden sector's state space.

**Statement of the theorem.** Under Axioms 1–3, the finiteness condition, and conditions (C1)–(C3), the embedded observer's reduced description of the visible sector is mathematically equivalent to unitarily evolving quantum mechanics. The remainder of Part I constitutes the proof. The conditions are shown to be not merely sufficient but *necessary* for P-indivisibility in any deterministic dilation (§3.3), establishing a full equivalence between quantum mechanics and embedded observation.

## 1.4 Partition-Relativity

Before proceeding to the proof, a structural consequence of the axioms is derived that will constrain the emergent description throughout.

**Lemma 1** (Partition-relativity)**.** *The emergent description of the visible sector is uniquely determined by the partition. Two observers whose causal access defines the same parti-*

*tion obtain the same reduced description. Any parameters of the emergent theory can depend only on quantities intrinsic to the partition — the geometric and thermodynamic properties of the boundary between visible and hidden sectors.*

*Proof.* By Axiom 1, the dynamics are deterministic: the Hamiltonian flow $\phi_t$ on $\Gamma_V \times \Gamma_H$ is a single, uniquely defined map. By Axiom 2, the partition $\Gamma = \Gamma_V \times \Gamma_H$ is fixed. By Axiom 3, the observer uses the Liouville measure $\mu$ on $\Gamma_H$ — the unique absolutely continuous measure preserved by the Hamiltonian flow (Liouville's theorem); any other smooth measure would evolve under the dynamics, introducing dependence on an arbitrary reference time, and singular invariant measures (e.g., those supported on periodic orbits) are excluded by Axiom 3's requirement that the observer assigns classical probability distributions over the full phase space. The marginalized transition probabilities

$$T_{ij}(t_2, t_1) = \int_{\Gamma_H} \delta_{x_j}[\pi_V(\phi_{t_2-t_1}(x_i, h))]\, d\mu(h) \tag{5}$$

are therefore uniquely determined by three inputs: the dynamics ($\phi_t$), the partition ($\Gamma_V, \Gamma_H$), and the measure ($\mu$). Since $\phi_t$ and $\mu$ are fixed by the axioms, the transition probabilities — and hence the complete stochastic process — depend only on the partition. If this process is P-indivisible (as shown in §2), the stochastic-quantum correspondence maps it to a quantum description. Since the stochastic process is unique, the quantum description is unique (up to unitary equivalence and gauge freedoms resolved below). In particular, any free parameters of the emergent theory — including the action scale $\hbar$ — can depend only on quantities intrinsic to the partition: the geometric and thermodynamic properties of the boundary between $\Gamma_V$ and $\Gamma_H$. □

**Why this matters.** Partition-relativity is not an additional postulate but a direct consequence of the axioms' deterministic and measure-theoretic content. It will play three roles in what follows: (i) it ensures that the emergent quantum mechanics is universal for all sub-$c$ observers sharing the same cosmological horizon (§4.1); (ii) it constrains the action scale $\hbar$ to depend only on quantities intrinsic to the partition, and specifically to its local *structure* rather than its volume, which together with dimensional analysis and thermal self-consistency determines the exact value $\hbar = c^3\epsilon^2/(4G)$ (§5.2); and (iii) it provides the logical bridge between the general theorem and the quantitative derivation: the same principle that guarantees uniqueness also restricts which physical quantities can appear in the emergent description.

# 2 EMERGENT STOCHASTICITY AND P-INDIVISIBILITY

## 2.1 Emergent Stochasticity

The total system (visible + hidden sectors) evolves deterministically (Axiom 1). Every phase-space point $(x, h) \in \Gamma_V \times \Gamma_H$ maps to exactly one future point under the Hamiltonian flow. No randomness exists at this level. But the visible sector alone — what the embedded observer accesses — is stochastic. The observer knows $x$ but not $h$. Different hidden-sector states $h_k$ compatible with the same $x$ send $x$ to different visible-sector futures $x'_k$. The observer must assign transition probabilities $T_{ij}(t_2, t_1)$ by marginalizing the deterministic flow over the hidden sector using the Liouville measure, as defined in §1.4. As established there, the Liouville measure is the unique absolutely continuous measure preserved by the Hamiltonian flow, ensuring that the transition probabilities — and hence the emergent description — are uniquely determined by the partition.

This architecture makes the framework a hidden variable theory: the randomness observed by the embedded observer is epistemic, arising from ignorance of hidden-sector degrees of freedom rather than fundamental indeterminacy. The consistency of this structure with Bell's theorem is addressed in §3.2.

## 2.2 The Slow-Bath Regime

The Markovian limit for reduced dynamics requires a strict timescale separation $\tau_B \ll \tau_S$ [4], where the environment decorrelates much faster than the system evolves. Condition (C2) inverts this hierarchy: $\tau_S \ll \tau_B$. The hidden sector evolves on timescales far exceeding those of any visible-sector process, so its correlations do not decorrelate between successive system transitions. The memory kernel of the reduced dynamics is effectively constant over all observationally relevant timescales.

A slow bath must be distinguished from a static external field. A genuinely static (decoupled) hidden sector would merely shift Hamiltonian parameters in $H_V$ without inducing information backflow. By condition (C1), the hidden sector is not decoupled: the interaction Hamiltonian $H_{\text{int}}$ is continuously active. During each visible-sector transition, $H_{\text{int}}$ imprints the identity of the transition onto the hidden-sector state. Because the hidden sector is slow — not static — these imprints persist without being thermally overwritten: the hidden sector's internal relaxation timescale ($\tau_B$) vastly exceeds the interval between successive visible-sector transitions ($\tau_S$). On subsequent transitions, the coupling reads back the stored correlations, producing transition probabilities that depend on the system's prior trajectory — the non-Markovian open-systems regime of Breuer and Petruccione [4].

## 2.3 P-Indivisibility

**Theorem 2** (P-indivisibility of embedded observation)**.** *Let $\mathcal{C}_V$ and $\mathcal{C}_H$ be finite sets with $|\mathcal{C}_V| \geq 2$, and let $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ be a bijection. Define the one-step transition matrix on $\mathcal{C}_V$ by marginalizing over $\mathcal{C}_H$ with respect to the uniform measure:*

$$T_{ij} = \frac{|\{h \in \mathcal{C}_H : \pi_V(\varphi(x_i, h)) = x_j\}|}{|\mathcal{C}_H|} \tag{6}$$

*and the $k$-step transition matrix $T_{ij}^{(k)}$ by the same construction applied to $\varphi^k$. If $T$ is not a permutation matrix — i.e., the marginalized dynamics is genuinely stochastic — then the stochastic process generated by iterating $\varphi$ is P-indivisible.*

*Proof.* The proof uses the total variation distance $d(p, q) = \frac{1}{2}\sum_k |p_k - q_k|$ between probability distributions. A stochastic process is P-divisible if and only if $d(p(t), q(t))$ is non-increasing in $t$ for all initial distributions $p, q$ [6, 7]. P-indivisibility is established by exhibiting a non-monotonicity.

*Step 1 (Recurrence).* Since $\varphi$ is a bijection on a finite set, it has finite order: there exists $N \geq 1$ such that $\varphi^N = \mathrm{id}$. Every state $(x_i, h)$ returns to itself after $N$ steps, so $T^{(N)} = I$. In particular, $d(\delta_i T^{(N)}, \delta_j T^{(N)}) = d(\delta_i, \delta_j) = 1$ for all $i \neq j$.

*Step 2 (Strict contraction at step 1).* Since $T$ is row-stochastic and not a permutation matrix, there exist distinct indices $i, j$ whose rows have overlapping support: some column $l$ satisfies $T_{il} > 0$ and $T_{jl} > 0$. (If no two rows had overlapping support, their supports would be pairwise disjoint, forcing each row to concentrate on a single column — a permutation matrix, contradicting the hypothesis.) For such $i, j$:

$$d(\delta_i T, \delta_j T) = \frac{1}{2}\sum_k |T_{ik} - T_{jk}| < \frac{1}{2}\sum_k (T_{ik} + T_{jk}) = 1 \tag{7}$$

The strict inequality holds because every column $l$ with overlapping support contributes $|T_{il} - T_{jl}| < T_{il} + T_{jl}$ to the sum.

*Step 3 (Non-monotonicity).* Steps 1 and 2 together give:

$$d(\delta_i T^{(1)}, \delta_j T^{(1)}) < 1 = d(\delta_i T^{(N)}, \delta_j T^{(N)}) \tag{8}$$

The total variation distance decreases from step 0 to step 1, then increases from step 1 to step $N$. This is non-monotonic: the process is P-indivisible. □

The theorem requires only two inputs: deterministic bijective dynamics (Axiom 1) and non-trivial coupling (condition C1, which ensures $T$ is not a permutation matrix). The finiteness condition guarantees the finite configuration space on which the theorem operates. The proof mechanism is information backflow: information about the initial

visible-sector state leaks to the hidden sector via the coupling (Step 2) and is later read back by the deterministic dynamics (Step 1), producing the non-monotonic trace distance that characterizes P-indivisibility.

**Continuous-time extension.** The theorem is stated for discrete-time dynamics, but the physical setup involves continuous-time Hamiltonian flow. The extension follows from continuity and Poincaré recurrence. The Hamiltonian flow $\varphi_t$ on the finite-dimensional phase space (finiteness condition) preserves the Liouville measure on compact energy surfaces. The transition matrix $T_{ij}(t)$ is continuous in $t$, with $T(0) = I$. Non-zero coupling (C1) ensures that $T(t)$ departs from the permutation class for $t > 0$, so that $d(\delta_i T(t), \delta_j T(t)) < 1$ for some pair $i, j$ and some $t > 0$. By Poincaré's recurrence theorem, for any $\delta > 0$, there exists $t_R > 0$ such that the set of hidden-sector states satisfying $\pi_V(\varphi_{t_R}(x_i, h)) = x_i$ has measure $> 1 - \delta$, giving $d(\delta_i T(t_R), \delta_j T(t_R)) > 1 - 2\delta$. For $\delta$ small enough, this exceeds $d(\delta_i T(t), \delta_j T(t))$, establishing non-monotonic trace distance and hence P-indivisibility for the continuous-time process.

**Convergence with prior work.** The theorem converges with several independent lines of work: Pechukas [5] showed that reduced dynamics of a correlated system need not preserve positivity; Rivas, Huelga, and Plenio [6, 7] formalized the connection between divisibility failure and information backflow; Pollock et al. [8] derived necessary and sufficient Markovianity conditions via the process tensor framework; and most directly, Strasberg and Esposito [9] demonstrated non-Markovian reduced dynamics precisely in the slow-bath regime of condition (C2).

**The physical role of conditions (C2) and (C3).** The theorem establishes P-indivisibility from (C1) and finiteness alone. Conditions (C2) and (C3) are not needed for the mathematical result but are essential for its physical force. (C2) ensures that information backflow occurs on timescales accessible to the observer, not merely at Poincaré recurrence times that may exceed the age of the universe. (C3) ensures that P-indivisibility persists at physical scale: if the hidden sector has insufficient capacity, successive imprints overwrite earlier ones — the hidden sector saturates, and the reduced dynamics cross over to effectively Markovian behavior. The conjunction of (C2) and (C3) guarantees that P-indivisibility is strong, persistent, and observationally relevant.

## 2.4 Explicit Construction: A Minimal Model

The following minimal model illustrates the theorem and makes its mechanism explicit.

**The model.** The visible sector has two states: $x \in \{0, 1\}$. The hidden sector has six states: $h \in \{1, 2, 3, 4, 5, 6\}$. The total phase space has 12 states $(x, h)$. The dynamics are a deterministic, invertible permutation $\sigma$ on these 12 states, defined as six disjoint

transpositions:

$$\sigma = (0{,}1 \leftrightarrow 1{,}1)\ (0{,}2 \leftrightarrow 1{,}2)\ (0{,}3 \leftrightarrow 0{,}4)\ (0{,}5 \leftrightarrow 0{,}6)\ (1{,}3 \leftrightarrow 1{,}4)\ (1{,}5 \leftrightarrow 1{,}6) \tag{9}$$

The first two transpositions couple the visible and hidden sectors — they change $x$ while preserving $h$, implementing condition (C1). The remaining four permute hidden-sector labels within each visible-sector value, representing the hidden sector's internal dynamics. Since $\sigma$ is a product of disjoint transpositions, $\sigma^2 = \mathrm{id}$: the hidden sector retains perfect correlations between successive transitions, satisfying condition (C2). Six hidden-sector states per visible-sector value ensures the hidden sector is not saturated by a single transition, satisfying condition (C3).

**Transition matrices.** At $t = 0$, the observer knows $x$ but not $h$, and assigns the uniform (Liouville) measure over the 6 hidden states. At $t = 1$ (one application of $\sigma$):

From $x = 0$: two of six hidden states ($h = 1, 2$) send $x$ to 1; four ($h = 3, 4, 5, 6$) leave $x$ at 0. From $x = 1$: two of six ($h = 1, 2$) send $x$ to 0; four leave $x$ at 1. The one-step transition matrix is:

$$T(1,0) = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix} \tag{10}$$

At $t = 2$ ($\sigma^2 = \mathrm{id}$): every state returns to itself, so $T(2,0) = I$.

A Markov chain with transition matrix $T(1,0)$ would predict $T(2,0)_{\text{Markov}} = T(1,0)^2 = \left(\begin{smallmatrix} 5/9 & 4/9 \\ 4/9 & 5/9 \end{smallmatrix}\right)$ — continued mixing. The actual dynamics show complete *un-mixing*: the system returns to its initial state with certainty. This is information backflow.

**P-divisibility test.** If the process were P-divisible, one could write $T(2,0) = \Lambda(2,1) \cdot T(1,0)$ with $\Lambda(2,1)$ a valid stochastic matrix (non-negative entries, rows summing to 1). Computing:

$$\Lambda(2,1) = T(2,0) \cdot [T(1,0)]^{-1} = I \cdot \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \tag{11}$$

The matrix $\Lambda(2,1)$ has negative entries ($-1$). No valid stochastic matrix can serve as the intermediate propagator. **The process is P-indivisible.** □

**The mechanism.** The hidden sector acts as a memory register. At step 1, the transpositions $(0,h) \leftrightarrow (1,h)$ for $h \in \{1,2\}$ flip $x$ while preserving $h$. After step 1, a state now at $x = 1$ with $h \in \{1,2\}$ carries the record "I was at $x = 0$." At step 2, the same transposition reads this record and flips $x$ back. The visible-sector dynamics exhibit information backflow: the system appears to "remember" its origin, which is impossible for a memoryless process. The model illustrates the mechanism in the idealized limit of perfect recurrence ($\sigma^2 = \mathrm{id}$, i.e., $\tau_B/\tau_S = 2$). The cosmological realization of Part II has $\tau_S/\tau_B \sim 10^{-32}$; the P-indivisibility is more extreme, not less, when the hidden sector's

memory is more persistent.

# 3 THE EMERGENCE OF QUANTUM MECHANICS

## 3.1 The Stochastic-Quantum Correspondence

By Barandes' stochastic-quantum correspondence [10, 11], any indivisible stochastic process on a finite configuration space of size $n$ can be embedded into a unitarily evolving quantum system on a Hilbert space of dimension $\leq n^3$. A stochastic process on $n$ configurations is characterized by its transition matrices $T_{ij}(t_2, t_1)$. If the process is P-divisible, these matrices can be factored through intermediate times and the system admits a classical description at all scales. If it is P-indivisible, no such factoring exists: the transition matrices at fine time resolution develop negative entries (as in §2.4), violating classicality. Barandes shows that the Stinespring dilation theorem then guarantees the existence of a Hilbert space and a unitary operator $U(t)$ such that $T_{ij}(t) = |U_{ij}(t)|^2$ — the transition probabilities of the stochastic process are exactly the Born-rule probabilities of the quantum system. Indivisibility is not merely *analogous* to quantum behavior; it is *mathematically equivalent* to it. The P-indivisible process established in §2.3 satisfies the correspondence's hypotheses directly.

**From phase space to configuration space.** The axioms define the dynamics on a phase space $\Gamma$ with coordinates $(q, p)$, while Barandes' correspondence operates on a configuration space. The connection is the marginalization itself: the transition probabilities $T_{ij}(t_2, t_1)$ defined in §2.1 are projections of the full phase-space flow onto the configuration sector of $\Gamma_V$, with the momenta — along with all hidden-sector degrees of freedom — absorbed into the marginalization over the Liouville measure. The resulting stochastic process lives on the discrete configuration space defined by the finiteness condition, where the correspondence applies directly.

Three features of quantum mechanics emerge from this correspondence. The Schrödinger equation is the unique time-evolution law, arising from the differentiability of the unitary in time. The Born rule is not an additional postulate but the equilibrium distribution of the indivisible stochastic process [10, 11]. The action scale $\hbar$ requires more careful treatment. The correspondence guarantees the existence of a unitary evolution operator $U(t)$, but $U$ is dimensionless; $\hbar$ enters only when one defines the Hamiltonian $\hat{H}(t) \equiv i\hbar\,\partial_t U \cdot U^\dagger$, converting the dimensionless rate of change into energy units. The correspondence itself is silent about the value of this conversion factor — it establishes that a quantum description exists for any P-indivisible process, but not which action scale it carries. What fixes $\hbar$ is the physical content of the partition: the emergent description must be unique for a given partition (§1.4), so the action scale can depend only on quantities intrinsic

to the partition boundary. Its value is derived for the cosmological realization in Part II (§5).

**Finite-dimensionality.** Barandes' proof is established for finite-dimensional configuration spaces. In the general theorem (Part I), the finiteness condition guarantees applicability directly: the finite minimal cell volume ensures that any finite spatial region contains at most a finite number of independent cells, and the causal partition (Axiom 2) restricts the visible sector to a finite total volume. In the cosmological application (Part II), the same guarantee comes from the partition geometry: the horizon's finite area bounds the number of coupled modes to $A/\epsilon^2$, ensuring the effective configuration space is finite-dimensional without requiring the total phase space to be finite. Calvo [12] has extended the correspondence to infinite-dimensional systems, relaxing the finite-dimensionality requirement, but the extension is not required here. (The $n^3$ upper bound on Hilbert space dimension is a worst-case embedding guarantee, not an estimate of the physical dimension. For the cosmological application with $n \sim S_{\mathrm{dS}} \sim 10^{122}$, the physical Hilbert space dimension is expected to be much smaller — plausibly $\sim n$ rather than $\sim n^3$ — since the trace-out produces a structured process with extensive symmetries that the generic bound does not exploit.)

## 3.2 Bell Inequality Violations

The framework is a hidden variable theory: quantum randomness arises epistemically from ignorance of hidden-sector degrees of freedom rather than fundamental indeterminacy. Bell's theorem [13] rules out *local* hidden variable theories — those satisfying the factorizability condition $P(a, b \mid x, y, \lambda) = P(a \mid x, \lambda) \cdot P(b \mid y, \lambda)$, where $\lambda$ denotes hidden variables with distribution $\rho(\lambda)$, and $x, y$ are measurement settings assumed statistically independent of $\lambda$. The present framework evades Bell's theorem not by violating measurement independence (superdeterminism) but by violating factorizability through a mechanism that is nonetheless *causally local* — no superluminal signaling occurs.

**The mechanism: non-factorizable joint dynamics from a shared causal past.** Consider two subsystems $Q$ and $R$ that interact locally at preparation time $t'$ and are subsequently separated to spacelike-distant measurement stations. In the stochastic-quantum framework [10, 11, 14], the preparation interaction creates a *joint* transition matrix $T_{QR}(t_2, t_1)$ on the composite configuration space $\mathcal{C}_Q \times \mathcal{C}_R$. Because the joint process is P-indivisible (§2.3), this transition matrix does not factorize:

$$T_{QR} \neq T_Q \otimes T_R \tag{12}$$

This non-factorizability *is* entanglement, expressed at the stochastic level. The correlations are encoded in the *form of the dynamical law itself* — in the structure of the joint transition matrix — rather than in a hidden variable $\lambda$ over which one could integrate and

apply screening-off. Since the process is indivisible, there are no well-defined intermediate conditional probabilities that would permit the factorization Bell's theorem requires.

Reichenbach's common-cause principle does not apply to indivisible processes, because the principle presupposes that conditioning on the complete common cause renders the effects independent — a decomposition that indivisibility explicitly forbids.

**Causal locality without Bell locality.** The Jarrett decomposition splits Bell's factorizability condition into *parameter independence* (PI) and *outcome independence* (OI). Quantum mechanics violates OI while preserving PI. The present framework reproduces this structure. Barandes, Hasan, and Kagan [14] formalize causal locality as a marginalization condition on the joint transition matrix: for subsystems $Q$ and $R$ at spacelike separation, $\sum_r T_{(q,r),(q_0,r_0)}(t) = T^{(Q)}_{q,q_0}(t)$ for all $r_0$ — marginalizing over $R$'s final configuration yields $Q$'s transition probabilities independent of $R$'s initial configuration. The joint transition matrix carries non-factorizable correlations from the shared preparation, but these correlations are accessible only through joint statistics, never through local marginals.

**The Tsirelson bound from indivisibility plus causal locality.** Barandes, Hasan, and Kagan [14] prove that the conjunction of indivisibility and causal locality restricts the maximum CHSH correlator to exactly Tsirelson's bound $|S| \leq 2\sqrt{2}$. Divisible (classical) stochastic processes satisfy $|S| \leq 2$, reproducing Bell's bound. PR-box correlations ($|S| = 4$) cannot be realized because the required transition matrices have no unitary square root. This result receives independent support from Le et al. [15], who prove the temporal analogue: *divisible* quantum dynamics satisfies the temporal Tsirelson bound, while *indivisible* dynamics can exceed it.

**Measurement independence and the distinction from superdeterminism.** The framework preserves measurement independence by construction: the transition probabilities $T_{ij}(t_2, t_1)$ are defined independently of the initial distribution, and measurement settings enter through which subsystem couplings are activated, not through pre-established correlations with the prepared state.

Superdeterministic theories evade Bell's theorem by violating this independence; the present framework instead violates outcome independence — the P-indivisible dynamics propagate preparation correlations forward as a single, non-factorizable transition. This distinction has empirical content: superdeterministic theories generically predict deviations from standard statistical mechanics, while the present framework predicts none.

This placement — violating factorizability (specifically outcome independence in the Jarrett decomposition) while preserving both measurement independence and parameter independence — is precisely the class Fine's theorem [24] permits: models that reproduce quantum correlations through non-factorizable joint dynamics rather than through conspiracy between hidden variables and measurement settings.

## 3.3 The Characterization Theorem: Necessity of the Conditions

The theorem of §1.3 establishes sufficiency: the axioms, finiteness, and conditions (C1)–(C3) guarantee quantum mechanics. This section proves the converse — that quantum mechanics *requires* these conditions — establishing a full equivalence between quantum mechanics and embedded observation.

**The logical structure.** Barandes' stochastic-quantum correspondence is already bidirectional: a stochastic process is P-indivisible if and only if it is mathematically equivalent to unitarily evolving quantum mechanics [10, 11]. The equivalence

$$\text{QM} \iff \text{P-indivisibility} \tag{13}$$

is established. What remains is to prove the converse of the other link:

$$\text{P-indivisibility} \implies \text{embedded observation under (C1)–(C3)} \tag{14}$$

The full equivalence QM $\iff$ embedded observation (under the axioms, finiteness, and (C1)–(C3)) then follows by transitivity.

**Dilation existence (necessity of the axioms).** Any stochastic process on a finite configuration space can be realized as the marginalization of a deterministic process on a larger state space: one enlarges the state space to encode the randomness as ignorance of auxiliary variables and recovers the original process by tracing them out. This is a standard construction in classical probability theory. The dilation furnishes a deterministic total system (Axiom 1), a partition into visible and hidden sectors (Axiom 2), and classical marginalization (Axiom 3). The three axioms are therefore necessary in the sense that any stochastic process — and hence any P-indivisible process — admits a deterministic embedding with a causal partition. The finiteness of the configuration space is inherited from the original process.

**Theorem 3** (C1 equivalence)**.** *Let $\mathcal{C}_V$ and $\mathcal{C}_H$ be finite sets with $|\mathcal{C}_V| \geq 2$, and let $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ be a bijection. Define the stochastic process on $\mathcal{C}_V$ by marginalizing over $\mathcal{C}_H$ with respect to the uniform measure. The process is P-indivisible if and only if (C1) holds — that is, if and only if the marginalized one-step transition matrix $T$ is not a permutation matrix.*

*Proof.* The forward direction (C1 $\implies$ P-indivisibility) is the theorem of §2.3. The reverse is proved by contraposition. If $T$ is a permutation matrix, then $T$ represents a deterministic map $\sigma : \mathcal{C}_V \to \mathcal{C}_V$ with $T_{ij} = \delta_{j,\sigma(i)}$. The $k$-step transition matrix is $T^{(k)}_{ij} = \delta_{j,\sigma^k(i)}$, also a permutation matrix for every $k$. The intermediate propagator $\Lambda(k_2, k_1) = T^{(k_2)} \cdot [T^{(k_1)}]^{-1} = T^{(k_2-k_1)}$ is a permutation matrix and hence a valid stochastic matrix (non-negative entries, rows summing to 1) for all $k_2 > k_1$. The process is P-

divisible. Contrapositively, P-indivisibility requires $T$ not to be a permutation matrix, which requires non-trivial coupling: $H_{\text{int}} \neq 0$. □

**Theorem 4** (C2 necessity: Markovian limit)**.** *Let $\varphi_{coup} : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ be a bijection implementing the visible-hidden coupling, and let $\varphi_H : \mathcal{C}_H \to \mathcal{C}_H$ be an ergodic permutation (a single cycle of length $|\mathcal{C}_H|$). Define the interleaved dynamics $\Phi_m = (\mathrm{id}_V \times \varphi_H^m) \circ \varphi_{coup}$, where $m$ represents the number of hidden-sector mixing steps between coupling events ($m$ is the ratio $\tau_S/\tau_B$, so large $m$ represents the fast-bath regime $\tau_B \ll \tau_S$). Let $T_m^{(k)}$ denote the $k$-step visible-sector transition matrix of $\Phi_m$. Then for $m$ coprime to $|\mathcal{C}_H|$, the process $\{T_m^{(k)}\}$ converges to a Markov chain as $|\mathcal{C}_H| \to \infty$ with the coupling structure held fixed. In the Markovian limit — when the hidden sector's state is independent of the visible sector's history before each coupling event — the reduced process is exactly P-divisible.*

*Proof.* The proof proceeds in two parts: first, that the idealized Markovian limit is P-divisible; second, that fast mixing drives the system to this limit.

*Part 1 (Markovian limit is P-divisible).* Consider the idealized process in which the hidden-sector state is drawn independently from the uniform measure on $\mathcal{C}_H$ before each coupling event, regardless of the visible sector's history. The one-step transition matrix is:

$$T_{ij} = \frac{|\{h \in \mathcal{C}_H : \pi_V(\varphi_{\text{coup}}(x_i, h)) = x_j\}|}{|\mathcal{C}_H|} \tag{15}$$

Since the hidden sector is re-randomized at each step, the transition probability at step $k$ depends only on the current visible state, not on the history. The $k$-step transition matrix is therefore $T^{(k)} = T^k$ (the $k$-th matrix power). The intermediate propagator is $\Lambda(k_2, k_1) = T^{k_2 - k_1}$, which is a stochastic matrix for all $k_2 > k_1$ (any power of a stochastic matrix is stochastic). The process is P-divisible.

*Part 2 (Fast mixing approaches the Markovian limit).* Since $\varphi_H$ is an ergodic permutation (a single cycle of length $|\mathcal{C}_H|$), the map $\varphi_H^m$ with $m$ coprime to $|\mathcal{C}_H|$ is also a single cycle of length $|\mathcal{C}_H|$. For any initial distribution $\mu$ on $\mathcal{C}_H$, repeated application of an ergodic permutation drives $\mu$ toward the uniform distribution in the Cesàro sense: the time-averaged distribution $\bar{\mu}_N = \frac{1}{N}\sum_{k=0}^{N-1}(\varphi_H^m)_*^k \mu$ converges to the uniform distribution $\mu_{\text{unif}}$ as $N \to \infty$.

In the interleaved dynamics, between the $(k-1)$-th and $k$-th coupling events, the hidden sector undergoes $m$ steps of $\varphi_H$. If $m$ is large relative to $|\mathcal{C}_H|$, the hidden-sector distribution — conditioned on the visible-sector state — approaches uniformity before the next coupling event. Formally: for any $\delta > 0$ and any initial distribution $\mu_0$ on $\mathcal{C}_H$, the total variation distance satisfies $d((\varphi_H^m)_*\mu_0, \mu_{\text{unif}}) < \delta$ whenever $m$ contains a full cycle (i.e., $m \geq |\mathcal{C}_H|$, by the ergodic theorem for finite groups). In the limit where the mixing is complete, the hidden-sector distribution before each coupling is exactly uniform and independent of the visible-sector history. The transition probabilities at each step are then exactly $T_{ij}$, independent of the past, and the process reduces to Part 1.

*Contrapositive (necessity of C2).* If the process is P-indivisible, Part 1 shows that the hidden sector's state cannot be independent of the visible sector's history at the moment of each coupling. The hidden sector must retain memory of past visible-sector transitions — correlations that have not been overwritten by internal mixing. Part 2 shows that fast mixing ($\tau_B \ll \tau_S$, large $m$) drives these correlations to zero. Therefore P-indivisibility requires that the hidden sector's mixing timescale exceed the coupling timescale: $\tau_S \ll \tau_B$, which is condition (C2). □

*Remark.* The theorem formalizes the physical distinction between a slow bath and a fast bath. A slow bath (C2 satisfied) retains the imprints of past visible-sector transitions and reads them back on subsequent steps — the mechanism underlying information backflow. A fast bath (C2 violated) erases those imprints through internal mixing before they can be read back, destroying the memory that P-indivisibility requires. The inverted timescale hierarchy is not an incidental feature of the sufficiency proof; it is the essential condition that enables the hidden sector to function as a persistent memory register.

**Theorem 5** (C3 necessity: information-theoretic capacity bound)**.** *Let $\varphi$ be a bijection on $\mathcal{C}_V \times \mathcal{C}_H$ with $|\mathcal{C}_V| = n$ and $|\mathcal{C}_H| = m$. Let $X_0, X_1, X_2, \ldots$ denote the visible-sector trajectory and $H_0, H_1, H_2, \ldots$ the corresponding hidden-sector trajectory under the deterministic dynamics. The non-Markovian mutual information — the information that the visible sector's history provides about its future evolution, beyond what is contained in its present state — satisfies:*

$$I(X_{<t}; X_{>t} \mid X_t) \leq \log_2 m \tag{16}$$

*In particular, the total non-Markovian information capacity of the reduced process is bounded by the logarithmic size of the hidden sector.*

*Proof.* The total system evolves deterministically: $X_{t+1}$ and $H_{t+1}$ are deterministic functions of $(X_t, H_t)$. Therefore the entire future trajectory $X_{>t} = (X_{t+1}, X_{t+2}, \ldots)$ is a deterministic function of $(X_t, H_t)$. By the data processing inequality for Markov chains: if $A \to B \to C$ is a Markov chain, then $I(A; C) \leq I(A; B)$. Conditioning on $X_t$:

$$X_{<t} \to H_t \to X_{>t} \quad \text{(conditioned on } X_t\text{)} \tag{17}$$

This is a Markov chain because, given $X_t$ and $H_t$, the future $X_{>t}$ is fully determined and independent of $X_{<t}$. Applying the data processing inequality:

$$I(X_{<t}; X_{>t} \mid X_t) \leq I(X_{<t}; H_t \mid X_t) \leq H(H_t \mid X_t) \leq \log_2 |\mathcal{C}_H| = \log_2 m \tag{18}$$

where the second inequality is a standard property of mutual information ($I(A; B) \leq H(B)$), and the third is the maximum entropy of a random variable on $m$ values. □

**Corollary 6** (Necessity of C3)**.** *P-indivisibility across all $n$ visible-sector configurations requires non-Markovian correlations spanning the full configuration space. The non-Markovian information required to sustain P-indivisibility across $n$ modes scales as $\Omega(\log n)$: distinguishing the non-Markovian evolution of $n$ visible-sector states requires at least $\log_2 n$ bits of hidden-sector memory. By the capacity bound, this requires $\log_2 m \geq \Omega(\log n)$, i.e., $m \geq n^\alpha$ for some $\alpha > 0$. The hidden sector must scale with the visible sector — precisely condition (C3).*

*Proof.* Suppose the visible sector has $n$ configurations and the reduced process is P-indivisible with information backflow affecting all $n$ configurations. The non-monotonicity of the trace distance (§2.3) requires that the hidden sector stores which of the $n$ visible-sector states the system previously occupied — or more generally, stores a trajectory-dependent record that affects the future transition probabilities of each visible-sector state differently. Storing a record that distinguishes $n$ possible states requires at least $\log_2 n$ bits. By the capacity bound theorem, the total non-Markovian information is at most $\log_2 m$. Therefore:

$$\log_2 m \geq \log_2 n \implies m \geq n \tag{19}$$

If the P-indivisibility is to persist over $K$ successive transitions (i.e., the hidden sector must simultaneously store information about $K$ distinct past events affecting $n$ configurations), the bound tightens to $\log_2 m \geq K \cdot f(n)$ for some function $f$ depending on the strength of the correlations, requiring $m/n$ to grow with the observation window. In particular, if the observation window includes $K \sim n$ transitions (as in any physically realistic scenario where the observer probes the visible sector repeatedly), the hidden sector cannot absorb these imprints without saturation unless $m/n$ is large. This is condition (C3).

*Contrapositive:* if $m$ is not sufficiently large relative to $n$, the $\log_2 m$ capacity bound chokes the non-Markovian information flow before it can sustain P-indivisibility across the full visible sector. The hidden sector saturates — successive imprints overwrite earlier ones — and the reduced dynamics on the affected configurations cross over to effective P-divisibility. □

*Remark.* The capacity bound is *information-theoretic* — it does not depend on the details of the bijection $\varphi$, only on the cardinalities of the two sectors. This universality is what makes C3 a necessary condition rather than a model-dependent feature: *any* deterministic dilation of a P-indivisible process on $n$ configurations must have a hidden sector of at least $n$ states, and a hidden sector much larger than $n$ for the P-indivisibility to persist under repeated observation. The bound is tight in the sense that the minimal model of §2.4 achieves P-indivisibility with $m = 6$ for $n = 2$, consistent with $m \geq n$.

**Characterization theorem (equivalence of quantum mechanics and embedded observation).** Let $\mathcal{C}_V$ be a finite configuration space with $|\mathcal{C}_V| \geq 2$, and let $\mathcal{S}$ be a

stochastic process on $\mathcal{C}_V$. The following are equivalent:

(i) $\mathcal{S}$ is mathematically equivalent to unitarily evolving quantum mechanics (via Barandes' stochastic-quantum correspondence [10, 11]).

(ii) $\mathcal{S}$ is P-indivisible.

(iii) $\mathcal{S}$ arises from the marginalization of a deterministic bijection $\varphi$ on $\mathcal{C}_V \times \mathcal{C}_H$ over a hidden sector $\mathcal{C}_H$, with:

- (C1) non-trivial coupling: the marginalized transition matrix $T$ is not a permutation matrix;
- (C2) slow-bath memory: the hidden sector retains correlations between successive coupling events (its mixing timescale exceeds the visible sector's dynamical timescale);
- (C3) sufficient capacity: $|\mathcal{C}_H| \geq |\mathcal{C}_V|$, with $|\mathcal{C}_H|/|\mathcal{C}_V|$ large enough that the hidden sector is not saturated by visible-sector interactions over the relevant observation window.

*Proof.* (i) $\iff$ (ii) is Barandes' stochastic-quantum correspondence [10, 11]. (iii) $\implies$ (ii) is the theorem of §2.3. (ii) $\implies$ (iii) is proved by the three necessity theorems above: dilation existence provides the deterministic embedding and the axioms; the C1 equivalence theorem establishes that the coupling must be non-trivial; the C2 necessity theorem establishes that the hidden sector must retain memory; and the C3 capacity bound establishes that the hidden sector must have sufficient dimensionality. □

**Interpretive consequence.** The characterization theorem closes the logical gap that a sufficiency-only result would leave open. Quantum mechanics is not merely *compatible with* the epistemic situation of an embedded observer — it is *equivalent to* it. Every quantum system can be understood as one in which an observer is embedded in a larger deterministic system with non-trivial, slowly decorrelating coupling to a hidden sector of sufficient capacity. Conversely, every such embedding necessarily produces quantum mechanics in the observer's reduced description.

The interpretive consequences of the theorem, its scope and limitations, relation to prior work, and the status of key assumptions are collected in §9 (Discussion), after the cosmological application and predictions have been developed.

# PART II: THE COSMOLOGICAL APPLICATION

## 4 THE COSMOLOGICAL HORIZON AS CAUSAL PARTITION

### 4.1 The Partition

General relativity defines the causal structure of spacetime. The cosmological horizon is the boundary beyond which no signal propagating at or below $c$ can reach the embedded observer. Every physically realizable measurement technique — electromagnetic, gravitational-wave, neutrino — propagates at or below $c$ and therefore encounters the same boundary. The horizon implements the causal partition of Axiom 2 for all sub-$c$ observers simultaneously: the visible sector $\Gamma_V$ is the interior of the cosmological horizon, and the hidden sector $\Gamma_H$ is everything beyond it.

This identification is not a modeling choice but a consequence of GR's causal structure. A hypothetical observer with access to a causal channel not constrained by the light cone would have a different partition, a different (or absent) hidden sector, and by the theorem's own logic, a different emergent description. No such channel is available within GR. The specific quantum mechanics observed is the quantum mechanics of light-cone-bounded observers; its universality is a consequence of the causal structure, not a postulate.

### 4.2 Verification of the Conditions

**Condition (C1): Non-zero coupling.** The cosmological horizon is not a static wall. Stress-energy conservation across the horizon enforces dynamical correlations between the visible and hidden sectors: matter and radiation cross the horizon continuously, and the horizon area responds to changes in the interior energy density via the Friedmann equation. The coupling is bidirectional. **Condition (C1) is satisfied.**

**Condition (C2): Slow-bath timescale separation.** The hidden sector's shortest correlation timescale is set by the Hubble time: $\tau_B \gtrsim 1/H \sim 10^{17}$ s, since modes beyond the horizon evolve on timescales no shorter than the rate at which the horizon itself changes. (Modes deeper in the hidden sector have even longer correlation times.) For any laboratory-scale process with characteristic frequency $\omega \gg H$, the visible sector's

dynamical timescale is $\tau_S \sim 1/\omega \sim 10^{-15}$ s or shorter. The ratio is:

$$\frac{\tau_S}{\tau_B} \sim 10^{-32} \tag{20}$$

The hidden sector cannot decorrelate between any pair of visible-sector events. The Markovian approximation fails categorically. **Condition (C2) is satisfied.**

**Condition (C3): Sufficient capacity.** The hidden sector has $N_{\text{modes}} = A/\epsilon^2$ independent degrees of freedom, where $A = 4\pi c^2/H^2$ is the horizon area and $\epsilon$ is the discreteness scale of the finiteness condition. For any $\epsilon$ at or above the Planck scale, the horizon area is cosmological ($A \sim 10^{52}$ m$^2$) and the visible sector involves at most $\sim 10^{80}$ baryons. No laboratory process can appreciably perturb the hidden sector's state. The hidden sector is never saturated. **Condition (C3) is satisfied.**

### 4.3 Application of the Theorem

All three axioms hold. All three conditions hold. The finiteness condition — which is an independent assumption in Part I — is here a consequence of the partition geometry: the cosmological horizon has finite area $A = 4\pi c^2/H^2$, and all visible-hidden coupling passes through this boundary. The number of modes that participate in the trace-out is bounded by $A/\epsilon^2$, which is finite for any $\epsilon > 0$. The effective stochastic process therefore lives on a finite-dimensional configuration space, as Barandes' correspondence requires, without any assumption that the total phase space beyond the horizon is itself finite.

The theorem of Part I applies: the embedded observer's reduced description of the visible sector is P-indivisible, and by Barandes' stochastic-quantum correspondence [10, 11], mathematically equivalent to unitarily evolving quantum mechanics with an action scale $\hbar$ determined by the local structural properties of the partition boundary.

*What this step requires:* Part I plus Barandes. The finiteness condition is provided by the partition geometry. No additional physics.

## 5 DERIVATION OF THE EMERGENT ACTION SCALE ($\hbar$)

### 5.1 The Classical Horizon Temperature

The theorem guarantees that $\hbar$ exists but does not compute its value. Partition-relativity (§1.4) constrains $\hbar$ to depend only on quantities intrinsic to the partition — the geometric and thermodynamic properties of the boundary between visible and hidden sectors. The required thermodynamic input is model-independent: it follows from Jacobson's thermodynamic derivation of Einstein's field equations [16].

Jacobson (1995) demonstrated that applying the thermodynamic relation $dE = T\,dS$ to local causal horizons yields Einstein's field equations. The same identity relates mass-energy to area via:

$$dE = \frac{c^2\kappa}{8\pi G}\,dA \tag{21}$$

The classical entropy density on the horizon is $\eta = 1/\epsilon^2$ — one degree of freedom per minimal cell (finiteness condition). Therefore $dS = dA/\epsilon^2$, and equating $dE = T_{\rm cl}\,dS$ fixes the classical temperature:

$$T_{\rm cl} = \frac{c^2\epsilon^2\kappa}{8\pi G k_B} \tag{22}$$

For the de Sitter horizon, the surface gravity is $\kappa = cH$, giving:

$$k_B T_{\rm cl} = \frac{c^3\epsilon^2 H}{8\pi G} \tag{23}$$

This temperature depends on $G$, $c$, $H$, $\epsilon$, and $k_B$ — no $\hbar$. It is the temperature of the hidden sector as a classical thermal reservoir, derived from the horizon geometry without reference to quantum mechanics.

## 5.2 The Emergent Action Scale

The stochastic-quantum correspondence maps the P-indivisible transition matrices to a unitary evolution $U(t)$, but leaves an enormous gauge freedom: the transition matrix constrains only $|U_{ij}|^2$, leaving the quantum phases — and hence the energy eigenvalues and the overall action scale — undetermined. This is the Schur–Hadamard gauge freedom inherent in the stochastic-to-quantum map. Partition-relativity (§1.4) constrains the resolution: the emergent quantum description is uniquely determined by the partition, so the action scale $\hbar$ can depend only on quantities intrinsic to the partition — the geometric and thermodynamic properties of the boundary between visible and hidden sectors.

The determination of $\hbar$ proceeds in four steps. Steps 1–3 establish the scaling; Step 4 fixes the prefactor.

**Step 1: Uniqueness from partition-relativity.** The lemma of §1.4 establishes that the emergent description is not merely *constrained* by the partition but *uniquely determined* by it. The three inputs — the dynamics $\phi_t$ (fixed by Axiom 1), the Liouville measure $\mu$ (fixed by Axiom 3), and the partition $\Gamma_V \times \Gamma_H$ (fixed by Axiom 2) — yield a single stochastic process, which maps to a single quantum description up to unitary equivalence. The action scale $\hbar$ is therefore not a free parameter to be fitted but a derived quantity: it is whatever value the unique correspondence assigns. There is no family of consistent quantum descriptions parameterized by different choices of $\hbar$.

**Step 2: The action scale is structural, not volumetric.** Among the quantities intrinsic to the cosmological partition, some characterize the *structure* of the coupling

across the boundary — the discreteness scale, the gravitational coupling, the speed of light — while others characterize the *volume* of the hidden sector — the horizon area $A$, the number of modes $N_{\text{modes}} = A/\epsilon^2$, or equivalently the Hubble parameter $H$ through $A = 4\pi c^2/H^2$. The surface gravity $\kappa$ occupies an intermediate position: it is a local geometric property of the horizon, but for de Sitter spacetime $\kappa = cH$, so it carries implicit $H$-dependence. Its classification is addressed quantitatively below (after Step 3).

The action scale $\hbar$ governs the conversion between the stochastic and unitary descriptions: it sets the rate at which the unitary $U(t)$ accumulates phase. This conversion is determined by the *character* of the P-indivisibility — how the hidden sector couples to the visible sector and imprints correlations — not by the total capacity of the hidden sector.

The distinction is analogous to a classical thermal reservoir: the temperature of a heat bath determines the Boltzmann factor $e^{-E/k_BT}$ regardless of the bath's total size; adding more degrees of freedom at the same temperature changes the heat capacity but not the thermal fluctuation spectrum. If $\hbar$ depended on $H$ through volumetric factors, observers at different cosmological epochs would obtain different action scales — different quantum mechanics entirely.

This argument is made precise by the following factorization result, combined with the D-gauge completeness theorem of §5.3.

**Lemma 7** (Volume-independence of transition probabilities)**.** *Let $(\mathcal{C}_V, \mathcal{C}_H, \varphi_t)$ be a deterministic system with transition matrix $T_{ij}(t)$ on $\mathcal{C}_V$. Decompose $\mathcal{C}_H = \mathcal{C}_B \times \mathcal{C}_D$ into boundary modes $\mathcal{C}_B$ and deep modes $\mathcal{C}_D$. If the visible-sector transition is independent of the deep modes on timescales up to $t_*$ — that is, $\pi_V(\varphi_t(x, h_B, h_D)) = \pi_V(\varphi_t(x, h_B, h'_D))$ for all $h_D, h'_D$ and all $t \le t_*$ — then $T_{ij}(t) = T^{(B)}_{ij}(t)$ for all $t \le t_*$, where $T^{(B)}$ is computed by marginalizing over $\mathcal{C}_B$ alone.*

*Proof.* For $t \le t_*$:

$$T_{ij}(t) = \frac{1}{|\mathcal{C}_B||\mathcal{C}_D|} \sum_{h_B, h_D} \mathbf{1}[\pi_V(\varphi_t(x_i, h_B, h_D)) = x_j] \tag{24}$$

By hypothesis, the indicator function is independent of $h_D$, so the sum over $\mathcal{C}_D$ contributes a factor of $|\mathcal{C}_D|$ that cancels the $1/|\mathcal{C}_D|$ prefactor:

$$T_{ij}(t) = \frac{1}{|\mathcal{C}_B|} \sum_{h_B} \mathbf{1}[\pi_V(\varphi_t(x_i, h_B)) = x_j] = T^{(B)}_{ij}(t) \tag{25}$$

□

The lemma extends to approximate independence: if the visible-sector transition depends on the deep modes for at most a fraction $\delta$ of configurations, then $|T_{ij}(t) - T^{(B)}_{ij}(t)| \le \delta$.

**Application to the cosmological partition.** The boundary modes are the $A/\epsilon^2$ degrees of freedom on the horizon surface; the deep modes are degrees of freedom at distances $\gg c/H$ beyond the horizon. Deep modes can influence the boundary only by propagating through the hidden sector, which takes time $\gtrsim \tau_B$ (C2). For all $t \ll \tau_B$, the lemma's condition is satisfied with $\delta \lesssim t/\tau_B \sim 10^{-32}$, and the transition probabilities are volume-independent. By the D-gauge theorem (§5.3), the transition probabilities determine all dimensionless physical content of the emergent quantum mechanics. The volume-independence of $\hbar$ follows: the process tensor is insensitive to deep modes, and the process tensor determines all physics.

**Step 3: Dimensional determination.** With volumetric quantities excluded, the partition-intrinsic quantities that can appear in $\hbar$ are restricted to those characterizing the local structure of the boundary: $c$ (defines the causal boundary), $G$ (enters through Jacobson's identity relating geometry to energy), and $\epsilon$ (sets the phase-space grain). The unique combination with dimensions of action is $c^3\epsilon^2/G$, giving:

$$\hbar = \beta \frac{c^3\epsilon^2}{G} \tag{26}$$

where $\beta$ is a dimensionless prefactor. Steps 1–3 establish that $\beta$ exists and is unique; they do not determine its value. The structural/volumetric distinction excludes $H$-dependence, and dimensional analysis selects the functional form, but a dimensionless constant cannot be fixed by dimensional analysis alone.

**The status of the surface gravity.** The surface gravity $\kappa$ is a local geometric property of the horizon and could in principle contribute to $\hbar$ as a structural quantity. Dimensional analysis settles the question: the most general combination of $c$, $G$, $\epsilon$, and $\kappa$ with dimensions of action is $(c^3\epsilon^2/G) \cdot (\kappa\epsilon/c^2)^e$ for any exponent $e$. For the de Sitter horizon, $\kappa\epsilon/c^2 = H\epsilon/c \sim Hl_p/c \sim 10^{-61}$. Any non-zero power of $e$ gives a correction suppressed by at least 61 orders of magnitude. The leading-order result $\hbar \propto c^3\epsilon^2/G$ is therefore robust: even if $\kappa$ enters the exact formula, its contribution is parametrically negligible.

**Step 4: Thermal self-consistency fixes the prefactor.** The derivation so far has produced two independent expressions for the thermal state of the partition boundary, one from each level of description.

*Before the trace-out* (§5.1), the classical horizon temperature is derived from Jacobson's thermodynamic identity and the entropy density $\eta = 1/\epsilon^2$:

$$k_B T_{\mathrm{cl}} = \frac{c^3\epsilon^2\kappa}{8\pi G} \tag{27}$$

*After the trace-out*, the emergent quantum mechanics is established by Part I with an action scale $\hbar$ whose scaling is $\hbar = \beta c^3\epsilon^2/G$ (Step 3) but whose prefactor $\beta$ is not

yet fixed. Quantum field theory on curved spacetime — which is now a *consequence* of the emergent description, not an independent input — predicts the Gibbons–Hawking temperature [17] for any horizon with surface gravity $\kappa$:

$$k_B T_{\mathrm{GH}} = \frac{\hbar\kappa}{2\pi c} \tag{28}$$

These two temperatures describe the same physical boundary. $T_{\mathrm{cl}}$ is the temperature of the hidden sector as a classical thermal reservoir, derived from the partition geometry. $T_{\mathrm{GH}}$ is the temperature that the emergent quantum description assigns to the same horizon. If $T_{\mathrm{cl}} \neq T_{\mathrm{GH}}$, the emergent description is internally inconsistent: it predicts a different thermal state for the very boundary that produced it.

Self-consistency requires $T_{\mathrm{cl}} = T_{\mathrm{GH}}$. For the de Sitter horizon ($\kappa = cH$):

$$\frac{c^3\epsilon^2 H}{8\pi G} = \frac{\hbar H}{2\pi} \tag{29}$$

The Hubble parameter $H$ cancels — confirming the structural/volumetric point, since the result is $H$-independent. Solving for $\hbar$:

$$\boxed{\hbar = \frac{c^3\epsilon^2}{4G}} \tag{30}$$

The prefactor is $\beta = 1/4$. No free parameter remains.

**Why this is not circular.** The argument uses the Gibbons–Hawking temperature, which is a result of quantum field theory on curved spacetime — the very theory whose emergence the theorem establishes. The logic is sequential, not circular:

1. Part I establishes that the emergent description *is* quantum mechanics, with *some* action scale $\hbar$ (existence, not value).

2. The emergent QM, applied to the de Sitter horizon, predicts $T_{\mathrm{GH}}(\hbar) = \hbar H/(2\pi k_B)$. This is a *consequence* of the emergent theory, expressed in terms of the unknown $\hbar$.

3. The classical temperature $T_{\mathrm{cl}}$ is derived independently in §5.1, with no $\hbar$.

4. Matching the two — requiring that the emergent description be consistent with the conditions that produced it — gives one equation in one unknown.

This is a *gap equation*: the same logical structure as self-consistent mean-field calculations in condensed matter physics, where the order parameter is determined by requiring the emergent collective description to reproduce the microscopic conditions that generated it. The Gibbons–Hawking temperature is not imported as an external postulate; it is a *prediction* of the emergent theory that must be consistent with the classical temperature that preceded the theory's emergence.

**Robustness of the thermal self-consistency condition.** The derivation above uses the Gibbons–Hawking formula, a result of QFT on curved spacetime — an emergent description that lives on a lattice with cutoff $\epsilon$. Two independent arguments establish that the self-consistency condition is exact to the accuracy needed.

*Primary argument (thermodynamic equilibrium).* The self-consistency condition admits a derivation that bypasses the Gibbons–Hawking formula entirely. The hidden sector is a thermal reservoir at $T_{\rm cl}$ (§5.1); the visible sector equilibrates with it (conditions C1–C3 ensure persistent thermal contact without saturation). In the emergent quantum description — finite-dimensional quantum mechanics, established by Part I — the equilibrium state of a system coupled to a thermal reservoir at temperature $T$ is the Gibbs state $\rho \propto e^{-H_{\rm eff}/k_B T}$. This is a theorem of quantum statistical mechanics requiring no continuum approximation. By the zeroth law, $T_Q = T_{\rm cl}$ exactly. The Gibbs state temperature depends on $\hbar$ through $\hat{H}_{\rm eff} = i\hbar\,\partial_t U \cdot U^\dagger$, giving a single equation in one unknown. The Gibbons–Hawking formula is then *recovered* as what this equation evaluates to for the de Sitter geometry — a prediction of the framework, not an input.

*Quantitative bound.* The Euclidean periodicity of the de Sitter metric — which determines the KMS temperature — is a topological property that the emergent lattice inherits exactly. Lattice corrections enter only as powers of $\epsilon H/c \sim 10^{-61}$, bounding the relative correction to $\beta = 1/4$ by $\mathcal{O}((\epsilon H/c)^2) \sim 10^{-122}$. This is the same trans-Planckian insensitivity established by Unruh's sonic analogue [28] and Fredenhagen and Haag's algebraic derivation [29].

The value $\hbar = c^3\epsilon^2/(4G)$ is the end product of a chain of constraints: partition-relativity fixes $\hbar$ uniquely (Step 1), the structural nature of the correspondence eliminates $H$-dependent factors (Step 2), dimensional analysis selects the functional form (Step 3), and thermal self-consistency fixes the prefactor (Step 4).

*What this step requires:* Partition-relativity (§1.4), the structural/volumetric distinction, Jacobson's thermodynamic identity, the Gibbons–Hawking temperature as a consequence of the emergent QM (Part I), and the self-consistency requirement that these describe the same boundary. No mode-structure assumptions, microphysical coupling model, or spectral density. The analysis of §5.3 clarifies why this step is necessary (dimensional obstruction) and confirms its sufficiency (D-gauge completeness).

## 5.3 Gauge-Fixing, the Dimensional Obstruction, and the Role of Thermal Self-Consistency

The stochastic-quantum correspondence maps the P-indivisible process to a unitary $U(t)$, but leaves the Schur–Hadamard gauge freedom: the transition probabilities $T_{ij} = |U_{ij}|^2$ do not uniquely determine the quantum phases. Two questions arise: (1) how much of this freedom can be resolved from the transition-probability data, and (2) can the action

scale $\hbar$ be determined without the thermal self-consistency of Step 4? The analysis below shows that the answers are "all of it, up to a physically trivial residual" and "no — for a structural reason."

**Process tensor uniqueness.** The P-indivisible stochastic process is characterized by the complete hierarchy of multi-time correlation functions — the *process tensor* $\mathcal{T}_{n:0}$ of Pollock et al. [8]. Every process tensor admits a unitary dilation. In the present framework, the dilation is not a free choice: the "environment" is the hidden sector in the thermal state $T_{\mathrm{cl}}$ (§5.1), and the total dynamics are the unique Hamiltonian flow (Axiom 1). The process tensor is therefore uniquely determined by the partition.

**The phase ambiguity resolution (Conceptual Bridge).** The central mathematical challenge in deriving a unique quantum description from classical probabilities is the Schur–Hadamard gauge freedom: transition probabilities $T_{ij} = |U_{ij}|^2$ strictly define the moduli of the unitary matrix elements but leave their quantum phases completely ambiguous. Left unresolved, this ambiguity would imply that a single stochastic process could map to a vast family of distinct quantum realities with entirely different interference patterns and energy eigenvalues. However, because the observer has access to continuous-time transition data — not just a single static snapshot — the dynamics over time rigidly lock the phases together. The theorem below demonstrates that for any generic Hamiltonian, matching the continuous time-evolution of the transition probabilities forces the quantum phases into a uniquely determined configuration. The only remaining freedom is a time-independent basis rephasing (the D-gauge) that has absolutely no observable physical consequences.

**Theorem 8** (D-gauge completeness)**.** *Let $U(t) = e^{-iHt}$ be a unitary propagator on an $n$-dimensional Hilbert space with a fixed configuration basis $\{|i\rangle\}$, where $H$ has non-degenerate eigenvalues $\{E_a\}$ and non-vanishing overlaps $|\langle i|a\rangle| > 0$ for all $i, a$. If $U'(t) = e^{-iH't}$ satisfies $|U'_{ij}(t)|^2 = |U_{ij}(t)|^2$ for all $i, j$ and all $t$, then $H' = DHD^\dagger$ for a time-independent diagonal unitary $D = \mathrm{diag}(e^{i\alpha_1}, \ldots, e^{i\alpha_n})$.*

*Proof.* Write $c_{ia} = \langle i|a\rangle$. Expanding in the energy eigenbasis:

$$|U_{ij}(t)|^2 = \sum_{a,b} c_{ia}\, c^*_{ja}\, c^*_{ib}\, c_{jb}\, e^{-i(E_a - E_b)t} \tag{31}$$

This is a Fourier series with frequencies $\omega_{ab} = E_a - E_b$. Non-degeneracy ensures the frequencies are distinct, so the Fourier coefficients $c_{ia}c^*_{ja}c^*_{ib}c_{jb}$ are uniquely determined. Matching these for $U'$ requires $c'_{ia}c'^*_{ja}c'^*_{ib}c'_{jb} = c_{ia}c^*_{ja}c^*_{ib}c_{jb}$ for all $i, j, a, b$. Setting $a = b$ and using $|c_{ia}| > 0$: the moduli $|c'_{ia}| = |c_{ia}|$ are fully determined. Writing $c'_{ia} = c_{ia}\, e^{i\delta_{ia}}$ and substituting into the general constraint:

$$e^{i(\delta_{ia} - \delta_{ja} + \delta_{jb} - \delta_{ib})} = 1 \quad \text{for all } i, j, a, b \tag{32}$$

The general solution is $\delta_{ia} = \alpha_i + \beta_a$ for arbitrary phases $\{\alpha_i\}$, $\{\beta_a\}$. The shift $c'_{ia} = c_{ia}\, e^{i(\alpha_i+\beta_a)}$ gives $|a'\rangle = e^{i\beta_a} D|a\rangle$ with $D = \mathrm{diag}(e^{i\alpha_i})$, so:

$$H' = \sum_a E_a |a'\rangle\langle a'| = DHD^\dagger \tag{33}$$

The $\beta_a$ cancel in the projectors and contribute only unobservable eigenstate rephasing. □

The D-gauge is physically trivial: $U \to DUD^\dagger$ corresponds to $|i\rangle \to e^{i\alpha_i}|i\rangle$, a time-independent basis rephasing that leaves all transition probabilities, energy-level spacings, interference patterns, and oscillation frequencies invariant.

*Non-degeneracy conditions.* The theorem requires non-degenerate eigenvalues (ensuring distinct Fourier frequencies) and non-vanishing overlaps $|\langle i|a\rangle| > 0$ (ensuring all phases are constrained). Both hold generically: the degenerate set has measure zero among Hermitian matrices, and condition (C1) ensures the effective Hamiltonian has non-trivial coupling to all configuration-basis states. The theorem applies except on a measure-zero set of fine-tuned systems.

**Lemma 9** (Cross-interval matching)**.** *Let $U(t_3, t_2)$ and $U(t_2, t_1)$ be propagators with all $|U_{ij}| > 0$, each determined up to D-gauge within its interval. Then the cross-interval transition probabilities $T_{ik}(t_3, t_1) = |\sum_j U_{ij}(t_3, t_2)U_{jk}(t_2, t_1)|^2$ fix the relative D-gauge between intervals up to a single global phase.*

*Proof.* Independent D-gauges introduce a phase mismatch $\phi_j = \alpha_j^{(1)} - \alpha_j^{(2)}$ at the junction. The requirement $|\sum_j e^{i\phi_j} U_{ij}U_{jk}|^2 = |\sum_j U_{ij}U_{jk}|^2$ for all $i, k$ yields, from the cross-terms:

$$\cos(\phi_j - \phi_l + \theta_{ijlk}) = \cos(\theta_{ijlk}) \quad \text{for all } i, j, l, k \tag{34}$$

where $\theta_{ijlk} = \arg(U_{ij}\bar{U}_{il}U_{lk}\bar{U}_{jk})$. Since $\theta_{ijlk}$ varies with $(i, k)$ under the non-degeneracy hypothesis, the only solution is $\phi_j - \phi_l = 0$ for all $j, l$: the mismatch reduces to a single global phase $\phi_j = \phi$. □

**Corollary (time-dependent D-gauge completeness).** Under condition (C2), $H_{\text{eff}}(t)$ is approximately time-independent on intervals $\tau_S \ll \Delta t \ll \tau_B$. The D-gauge theorem applies within each interval; the cross-interval matching lemma — which is exact and algebraic — fixes the relative gauge at each junction up to a global phase. In the continuum limit $\Delta t \to 0$, the within-interval approximation error vanishes ($\mathcal{O}(\Delta t/\tau_B) \to 0$), and the chain of global phases becomes a smooth function $e^{i\theta(t)}$ — a time-dependent global phase that changes no transition probability, density matrix, or expectation value.

**Consequence.** The continuous-time transition probabilities $\{T_{ij}(t)\}_{i,j,t}$ determine the unitary propagator — for both time-independent and time-dependent effective gen-

erators — up to physically irrelevant rephasing. All gauge-invariant predictions of the emergent quantum mechanics are fixed by the process tensor alone.

**The dimensional obstruction and the role of Step 4.** The preceding results resolve all *dimensionless* ambiguity but leave $\hbar$ untouched — and this is a structural feature, not a gap. The unitary $U(t)$ is dimensionless; $\hbar$ enters only when defining $\hat{H} = i\hbar\,\partial_t U \cdot U^\dagger$, converting dimensionless rates into energy units. Different values of $\hbar$ yield different energies but the *same* $U(t)$ and transition probabilities. No quantity of dimensionless probability data can fix a dimensionful constant.

Step 4 provides the necessary and minimal dimensionful input: the thermal self-consistency condition $T_{\rm cl} = T_{\rm GH}(\hbar)$ connects the dimensionless oscillation frequencies to a physical temperature derived from the classical partition geometry (§5.1). This is not a crutch awaiting replacement but the unique completion of the dimensionless-to-dimensionful bridge. The process tensor determines all physical content (phase uniqueness); Step 4 converts frequencies to energies (scale determination). No other input is needed, and no alternative dimensionful equation would give a different $\hbar$, because $T_{\rm cl}$ is itself uniquely determined by the partition geometry.

*Status: Phase uniqueness established (D-gauge theorem + cross-interval matching + (C2)). Volume-independence established (volume-independence lemma + (C3)). Scale determination via thermodynamic equilibrium (finite-dimensional, no continuum input), with lattice corrections bounded by $\mathcal{O}(10^{-122})$. No remaining gap in the derivation of $\hbar$.*

# 6 SELF-CONSISTENCY AND THE DISCRETENESS SCALE

Before proceeding, it is crucial to reiterate that the classical temperature $T_{\rm cl}$ and the emergent quantum temperature $T_{\rm GH}$ were derived from completely independent starting points before being equated. The value $\hbar = c^3\epsilon^2/(4G)$ is established by the chain of constraints in §5.2: partition-relativity, the structural/volumetric distinction, dimensional analysis, and thermal self-consistency. Rearranging:

$$\epsilon^2 = \frac{4\hbar G}{c^3} = 4\,l_p^2 \tag{35}$$

The framework predicts $\epsilon = 2\,l_p$ exactly. The geometric discreteness scale is twice the Planck length — a specific quantitative prediction, not an order-of-magnitude estimate.

This value is necessary for self-consistency. If $\epsilon^2 \ll l_p^2$, sub-Planckian subcells would be dynamically active yet unresolvable by the emergent description, creating a second trace-out with its own noise intensity and breaking the single-valuedness of $\hbar$. If $\epsilon^2 \gg l_p^2$, the emergent description would assign distinct quantum states to configurations that are

physically identical in the substratum. The regime $\epsilon = \mathcal{O}(l_p)$ is the only self-consistent one; thermal self-consistency selects $\epsilon = 2\,l_p$ within it.

With $\epsilon = 2\,l_p$, the number of independent modes on the cosmological horizon is:

$$N_{\text{modes}} = \frac{A}{\epsilon^2} = \frac{A}{4\,l_p^2} = S_{\text{dS}} \tag{36}$$

where $S_{\text{dS}} = A/(4\,l_p^2)$ is the Bekenstein–Hawking entropy of the de Sitter horizon. The factor of 1/4 in the Bekenstein–Hawking formula — historically introduced as a proportionality constant — is here derived as a consequence of the partition geometry and the thermal self-consistency condition: each minimal cell of area $\epsilon^2 = 4\,l_p^2$ on the horizon contributes one unit of entropy, and $A/(4\,l_p^2) = A/\epsilon^2$ recovers the standard formula exactly.

**Consistency check.** After the emergence of quantum mechanics and the identification $\epsilon = 2\,l_p$, the classical horizon temperature becomes $T_{\text{cl}} = c^3\epsilon^2\kappa/(8\pi G k_B) = c^3(4\,l_p^2)\kappa/(8\pi G k_B) = \hbar\kappa/(2\pi c k_B)$ — the standard Gibbons–Hawking temperature [17] exactly. This recovery is guaranteed by construction (it is the self-consistency condition of Step 4), but verifying it confirms the algebraic consistency of the derivation.

# 7 DISSOLUTION OF THE COSMOLOGICAL CONSTANT PROBLEM

## 7.1 The Two Levels of Description

The cosmological constant problem [2, 3, 18] — the $10^{122}$-fold discrepancy between the QFT prediction for vacuum energy density and the observed value — is often called the worst prediction in physics. Its resolution turns on recognizing that the classical substratum and the emergent quantum description give different — and independently self-consistent — answers to the same question.

**From finite-dimensional QM to QFT.** The theorem of Part I delivers quantum mechanics on a finite-dimensional Hilbert space. The connection to quantum field theory proceeds through three standard steps. First, the finiteness condition discretizes the visible sector into $N = A/\epsilon^2$ cells of area $\epsilon^2$, each coupled to the hidden sector through the partition boundary. Second, the causal structure of the classical substratum restricts which cells interact on any given timescale, and the emergent quantum description inherits this locality. Third, spatial locality implies the emergent Hilbert space decomposes as $\mathcal{H} = \bigotimes_k \mathcal{H}_k$ — a lattice-regularized QFT with a built-in ultraviolet cutoff at $\epsilon = 2\,l_p$. The zero-point energy $\frac{1}{2}\hbar\omega$ per mode and the resulting QFT vacuum energy density are properties of this emergent description, not of the classical substratum. (The tensor

product decomposition is standard for spatially local Hamiltonian systems on a lattice; a rigorous derivation from the axioms requires showing that the Barandes correspondence preserves spatial locality — expected to hold generically but not proved here.)

**The classical substratum** (what geometric measurements probe): The cosmological horizon has a classical thermal equilibrium. By the Friedmann equation, $H^2 = (8\pi G/3)\rho$, the horizon area $A = 4\pi c^2/H^2$ adjusts self-consistently so that the mean energy density matches the critical density $\rho_{\text{crit}} = 3H^2c^2/(8\pi G)$. The classical energy per mode is $k_B T_H$. There is no zero-point energy and no discrepancy. The total vacuum energy density is at the critical scale $\rho \sim H^2/G \sim 10^{-9}$ J/m$^3$ — precisely what is observed.

**The emergent QFT** (what local quantum measurements probe): Quantum field theory assigns a zero-point energy of $\frac{1}{2}\hbar\omega$ per mode. Summing to the Planck cutoff yields $\rho_{\text{QFT}} \sim M_{\text{Pl}}^4 \sim 10^{113}$ J/m$^3$.

## 7.2 Why Gravity Sees the Classical Value

Spacetime geometry is part of the classical substratum (Axiom 2): the metric tensor is defined on the fundamental phase space and evolves via Einstein's field equations *before* the trace-out that produces quantum mechanics. The stress-energy tensor that sources the Einstein equations is the classical stress-energy of the total microstate, not the expectation value of an emergent quantum operator. The $10^{113}$ J/m$^3$ zero-point energy exists in the observer's informational ledger — it is a consequence of re-describing the classical noise as quantum fluctuations — but it does not appear in the stress-energy tensor that governs the geometry. The semiclassical gravity equation $G_{\mu\nu} = 8\pi G\langle\hat{T}_{\mu\nu}\rangle$ is itself an emergent approximation, valid when the quantum description is treated as fundamental. Within the present framework, where the quantum description is derived, the gravitational field equations operate at the classical level and never encounter the zero-point sum.

**Ontological commitment and its consequences.** The framework makes a specific ontological commitment: classical spacetime geometry is fundamental; quantum mechanics is emergent. This is not a free choice but follows from the logical structure of the derivation. The causal partition that produces quantum mechanics is defined by null geodesics of the metric (§4.1); the metric must therefore exist prior to the partition, and hence prior to the quantum description the partition generates. Reversing this ordering — treating the metric as emergent from quantum mechanics — would make the partition definition circular. This logical priority is opposite to programs that treat the metric as emergent (canonical quantum gravity, asymptotic safety, string-theoretic emergent spacetime), and the ordering question has observable consequences.

The cosmological constant problem is one such consequence. Any framework in which the quantum description is logically prior to the metric will necessarily treat the zero-

point sum as a source term in the gravitational field equations, because the stress-energy tensor is constructed from quantum operators. The $10^{122}$ discrepancy is then a real problem requiring cancellation or fine-tuning. In the present framework, where the metric is logically prior, the zero-point sum is an artifact of the emergent description and never enters the stress-energy tensor. The dissolution of the CC problem is therefore a direct prediction of the framework's central commitment — not a separate result, and not an ad hoc stipulation, but a consequence of taking seriously the derivation's own logical structure.

**Quantum corrections to gravity.** The claim is not that gravity is immune to quantum effects. The emergent quantum description does feed back into the gravitational dynamics through state-level quantities: §8.1 derives dark energy evolution in Running Vacuum form precisely because the emergent vacuum energy — a state-level property of the quantum description — depends on the Hubble parameter through the hidden sector's volume. This $H$-dependent running of $\Lambda_{\text{eff}}$ *is* a quantum correction to gravity.

What the framework excludes is the zero-point sum — a property of the correspondence itself rather than of any particular state — as a gravitational source. The structural/volumetric distinction of §5.2 applies here too: the action scale $\hbar$ (structural) does not gravitate; the vacuum energy (volumetric, state-dependent) does, but at the classical scale $\rho \sim H^2/G$ rather than the QFT scale $\rho \sim M_{\text{Pl}}^4$.

## 7.3 The Structural Origin of the $10^{122}$ Discrepancy

The $10^{122}$ factor has a precise structural interpretation. Summing the emergent zero-point energies to the Planck cutoff:

$$\rho_{\text{QFT}} \sim \int_0^{\omega_{\text{Pl}}} \frac{1}{2}\hbar\omega \cdot g(\omega)\, d\omega \sim M_{\text{Pl}}^4 \tag{37}$$

The ratio to the classical result is:

$$\frac{\rho_{\text{QFT}}}{\rho_{\text{classical}}} \sim \frac{M_{\text{Pl}}^4}{H^2/G} = S_{\text{dS}} \tag{38}$$

The discrepancy equals $S_{\text{dS}} = A/(4\, l_p^2)$ — the Bekenstein–Hawking entropy of the cosmological horizon, now identified as the number of independent modes on the partition boundary ($N_{\text{modes}} = A/\epsilon^2 = S_{\text{dS}}$). This is the information compression ratio of the quantum description: the number of hidden-sector degrees of freedom that the trace-out compresses into the emergent quantum state. The "worst prediction in physics" is the entropy of the observer's blind spot — a category error, not a fine-tuning failure.

That this discrepancy cannot be resolved from within is confirmed by Wolpert's (2008) limits of inference [19]: any physical subsystem whose state is a deterministic,

many-to-one function of the total configuration is subject to absolute inferential limits. Both geometric measurements (classical substratum) and local particle measurements (emergent QFT) are faithful to their respective descriptions; no embedded observer can determine from within which is more fundamental. Meanwhile, the observed value $\rho_\Lambda \sim 6 \times 10^{-10}$ J/m$^3$ is inferred from the expansion history — a classical geometric measurement reflecting $\rho \sim H^2/G \sim \rho_{\text{crit}}$, with $\rho_\Lambda/\rho_{\text{crit}} \approx 0.7$. The vacuum component is the residual energy density after matter has diluted, its value set by the current epoch's horizon geometry. No fine-tuning is required.

# 8 PREDICTIONS AND TESTS

## 8.1 Dark Energy Evolution in Running Vacuum Form

The action scale $\hbar$ depends only on structural properties of the partition boundary and is independent of $H$ (§5.2). But the emergent vacuum energy is a state-level quantity, not a property of the correspondence itself, and it does depend on the hidden sector's volume. The horizon area $A = 4\pi c^2/H^2$ evolves with the Hubble parameter. Since the hidden sector's dimensionality $N_{\text{modes}} = A/\epsilon^2$ depends on $H$, the emergent vacuum energy inherits a dependence on $H$. Expanding $\Lambda_{\text{eff}}(H)$ around the present epoch:

$$\Lambda_{\text{eff}}(H) = \Lambda_0 + \frac{3\nu_{\text{OI}}}{8\pi G}(H^2 - H_0^2) + \mathcal{O}((H^2 - H_0^2)^2) \tag{39}$$

where $\nu_{\text{OI}}$ is a dimensionless coefficient encoding the rate at which the hidden-sector dimensionality responds to changes in $H$. This is precisely the form of the Running Vacuum Model (RVM) [20, 21]. The framework constrains $\nu_{\text{OI}}$ in two ways without computing its exact value.

*Sign.* The framework requires $\nu_{\text{OI}} > 0$: as $H$ decreases, the horizon area grows, the hidden sector gains capacity, and the effective cosmological constant decreases. This sign prediction is unambiguous — it follows directly from the monotonic relationship between $H$ and $N_{\text{modes}}$.

*Magnitude.* Condition (C3) requires $N_H/N_V \gg 1$: the hidden sector must have vastly more degrees of freedom than the visible sector, so that visible-sector transitions do not appreciably perturb the hidden-sector state. This means the fractional change in the emergent description per unit change in $N_{\text{modes}}$ must be small — if each added mode materially altered the trace-out, the hidden sector would be effectively saturated and (C3) would be violated. The coefficient $\nu_{\text{OI}}$ measures precisely this fractional response, so (C3) requires $\nu_{\text{OI}} \ll 1$. For comparison, Solà Peracaula, Gómez-Valent, and de Cruz Pérez [21] fit the RVM to CMB, BAO, and SNIa data and consistently obtain $\nu_{\text{eff}} \sim \mathcal{O}(10^{-3})$, with the ΛCDM value $\nu = 0$ disfavored at 3–4$\sigma$ in several dataset combinations. The frame-

work's prediction — $\nu_{\rm OI} > 0$ and $\nu_{\rm OI} \ll 1$ — is consistent with this phenomenological range, though pinning the value to $\mathcal{O}(10^{-3})$ specifically requires the non-perturbative calculation identified as the principal open computational problem.

DESI's 2024–2025 data releases [22, 30] report evidence for evolving dark energy at $2.8\sigma$–$4.2\sigma$, with $w_0 > -1$ and $w_a < 0$ — the qualitative signature the RVM predicts. Direct RVM fits to the latest data have now been performed: de Cruz Pérez, Gómez-Valent, and Solà Peracaula [31] fit the RVM and several of its variants to DESI DR2 BAO combined with Planck PR4 CMB and both Pantheon+ and DES-Y5 supernovae, finding significant evidence for dynamical vacuum energy at $2.7\sigma$–$3.1\sigma$ depending on the model variant and dataset combination, with the ΛCDM value $\nu = 0$ disfavored at levels comparable to the generic $w_0w_a$CDM parameterization. These results are consistent with the framework's prediction of $\nu_{\rm OI} > 0$ and $\nu_{\rm OI} \ll 1$. The outstanding discriminating test is whether the one-parameter RVM form $(\Lambda_0 + \nu H^2)$ fits the data as well as or better than the two-parameter $w_0w_a$ model — a result that would favor the specific functional form this framework predicts over generic dark energy evolution. A non-perturbative calculation of $\nu_{\rm OI}$ from the horizon's spectral response to changes in $H$ would convert the present qualitative consistency into a quantitative test.

## 8.2 Gravitational Wave Echoes

A black hole horizon represents a regime where the causal partition differs from the cosmological one. For an observer near the horizon, degrees of freedom that are part of the cosmological visible sector become locally inaccessible — reclassified from visible to hidden sector as they cross the horizon. The effective partition shifts, and the emergent description must shift accordingly.

At a proper distance $\sim \epsilon$ outside a black hole horizon $r_h$, the local wavelength of an infalling mode reaches $\epsilon$ — the discreteness floor. The mode has no kinematically accessible states at shorter wavelengths and must scatter back. The predicted time delay is:

$$\Delta t_{\rm echo} \approx \frac{r_h}{c} \ln\left(\frac{r_h}{2\,l_p}\right) \tag{40}$$

For a $30M_\odot$ remnant with $\epsilon = 2\,l_p$, the logarithmic dependence on $\epsilon$ makes this prediction insensitive to the exact value: replacing $l_p$ with $2\,l_p$ shifts the delay by $\ln 2 \cdot r_h/c \approx 0.1$ ms, well within the $\sim 54$ ms timescale [23]. The exact reflection amplitude requires non-perturbative calculation. The echo signal, if detected, would constitute direct evidence that the discreteness scale $\epsilon$ identified by the self-consistency argument (§6) has observable consequences at horizons other than the cosmological one — a regime where the partition-relativity of the emergent description (§9.2) becomes empirically accessible.

## 8.3 Conjunction as Discriminant

The conjunction of confirmed dark energy evolution *and* detected GW echoes would uniquely favor an information-theoretic origin over standard QFT-derived RVMs, which predict $\Lambda(H)$ running but have no mechanism for discreteness-scale echoes. Each prediction alone is compatible with other frameworks; their joint confirmation would be distinctive.

# 9 DISCUSSION

## 9.1 Interpretive Consequences

**A note on "hidden sector" in what follows.** The general theorem applies to any partition satisfying the axioms, and the interpretive consequences below are stated in that generality. In the cosmological application of Part II, the hidden sector is specifically the trans-horizon region; the degrees of freedom involved in quantum experiments — photons, electrons, slits, detectors — are all visible-sector objects. Their quantum behavior is not produced by a separate marginalization over nearby modes but is a downstream consequence of the single cosmological trace-out: the emergent quantum mechanics, once established by the theorem, governs all visible-sector dynamics, including the local experiments discussed here.

The theorem resolves several foundational puzzles as direct consequences rather than additional postulates.

In the double-slit experiment, the particle traverses a single slit at the level of the deterministic substratum — it has a definite trajectory in the total phase space. The interference pattern arises because the observer does not have access to the total phase space. The transition matrix $T_{ij}(t_2, t_1)$ encodes the boundary conditions of the full experimental geometry, including the number of open slits. Opening or closing the second slit changes those boundary conditions, which changes the transition probabilities and hence the distribution of detection events. What appears as "wave-like" behavior is the trace-out-induced dynamics responding to a change in the visible-sector geometry. The which-path story follows the same logic: a detector at one slit couples the particle's trajectory to additional visible-sector degrees of freedom, altering the transition matrix and eliminating the interference terms.

In Wigner's friend scenarios, the Friend's measurement produces a definite outcome: the indivisible dynamics of the total system (Friend + measured system + hidden sector) evolve to a single state. Wigner, who lacks access to the lab's internal degrees of freedom, must trace them out — producing a superposition assignment that reflects his epistemic deficit, not the Friend's physical state. There is no contradiction because the

two descriptions correspond to different partitions, and partition-relativity (§1.4) permits different partitions to yield different reduced descriptions.

The Everettian measure problem dissolves because the total system evolves deterministically as a single reality — the Schrödinger equation is a mandatory compression algorithm for an observer who cannot track hidden-sector degrees of freedom, and "branches" are features of the compressed description, not of the underlying dynamics. The question "why does the Born rule give the right probabilities across branches?" is replaced by the answer that the Born rule *is* the equilibrium distribution of the indivisible stochastic process, derived rather than postulated.

## 9.2 Scope of the Theorem

**What the theorem says.** The characterization theorem (§3.3) establishes a full equivalence: a stochastic process on a finite configuration space is quantum mechanics if and only if it arises from embedded observation under conditions (C1)–(C3). The sufficiency direction — that the axioms, finiteness, and (C1)–(C3) produce quantum mechanics — is the constructive result. The necessity direction — that any quantum system *requires* non-trivial coupling, slow-bath memory, and sufficient hidden-sector capacity when realized as a deterministic dilation — is what distinguishes the framework from programs that merely *permit* an epistemic interpretation of quantum mechanics. The Schrödinger equation, Born rule, and Bell inequality violations are structural consequences of the equivalence, not independent postulates.

**What the theorem does not say.** It does not identify which physical systems satisfy the conditions. That is an empirical question. It does not derive the numerical value of $\hbar$ for any specific system — only that an action scale set by the trace-out noise necessarily exists. These questions are addressed for a specific physical realization in Part II.

**Partition-relativity: physical implications.** The lemma of §1.4 established that the emergent description is uniquely determined by the partition. The converse deserves explicit statement: *different* partitions produce *different* reduced descriptions. Different divisions of the total phase space into visible and hidden sectors yield different stochastic processes, potentially with different action scales or, if the conditions (C1)–(C3) are not met for a given partition, no quantum description at all. The equivalence does not say "quantum mechanics is the unique description of reality." It says "quantum mechanics and embedded observation under (C1)–(C3) are the same thing" — any system exhibiting one necessarily exhibits the other.

This is a conditional prediction, not a claim that distinct partitions are generically realized. "Embedded observers cannot access the complete state" is entirely consistent with all observers sharing the *same* inaccessible region — in which case the emergent

description is universal, not relative. Partition-relativity becomes physically significant only when distinct observers have distinct causal boundaries. The physical implications of this for our universe — where the causal partition is defined by null geodesics and all sub-$c$ observers share the same horizon — are developed in Part II (§4.1).

## 9.3 Dependence on Barandes' Correspondence

The theorem rests on Barandes' stochastic-quantum correspondence as its mathematical engine.

**What the correspondence establishes and does not.** Barandes [10, 11] proves that any indivisible stochastic process on a finite configuration space of size $n$ can be embedded into a unitarily evolving quantum system on a Hilbert space of dimension $\leq n^3$, and conversely that every quantum system can be understood as such a process. The correspondence is bidirectional and exact: it reproduces all first-order conditional probabilities — and hence all empirical predictions of quantum mechanics, including interference, entanglement, and decoherence — from indivisible stochastic dynamics on a classical configuration space. The Born rule emerges as the equilibrium distribution rather than an independent postulate. The proof rests on the Stinespring dilation theorem and standard linear algebra; it has been published in a peer-reviewed venue [11] and corroborated by independent follow-up work: Calvo [12] extended it to infinite-dimensional systems, Pimenta [25] provided a complete geometric characterization of divisible and indivisible dynamics within the correspondence's framework, and Doukas [26] proved several key structural properties while clarifying the conditions under which effective Markovian behavior can emerge from non-Markovian foundations. Barandes, Hasan, and Kagan [14] derived the Tsirelson bound from indivisibility plus causal locality. The higher-order conditional probabilities — the multi-time statistics beyond two-point transition matrices — are not uniquely fixed by the correspondence; different choices yield the same first-order predictions and hence the same empirical content. The present theorem is not affected by this ambiguity: partition-relativity (§1.4) uniquely determines the stochastic process, including its multi-time structure, from the physical content of the partition (§5.3).

**Where the correspondence enters.** The dependence is load-bearing but circumscribed. Partition-relativity, emergent stochasticity, and P-indivisibility (§§1.4, 2) do not invoke Barandes at all — they follow from the axioms and conditions alone. The correspondence enters at a single step: the map from P-indivisible stochastic process to quantum mechanics (§3.1). Everything before that step — including the result that the embedded observer cannot see classical physics and that information backflow produces non-factorizable correlations — is independent of the correspondence.

**Open questions.** The correspondence is a 2023 result that has not yet undergone decades of community scrutiny. While the core proof is mathematically straightforward

(Stinespring dilation plus linear algebra) and has been independently corroborated, the broader implications are still being explored. A restriction or refinement of the correspondence that narrowed the class of admissible indivisible processes could, in principle, affect the theorem's scope. The present framework's exposure to this risk is mitigated by the fact that the cosmological trace-out produces a highly structured process — not a generic indivisible process — and the correspondence's hypotheses are satisfied directly.

## 9.4 The Status of the Discreteness Scale

The discreteness scale $\epsilon$ warrants explicit discussion, because in a framework that derives quantum mechanics from classical ingredients, introducing a fundamental length scale may appear to smuggle in a quantum assumption.

It does not. The finiteness condition (§1.2) requires that the effective configuration space be finite-dimensional — a condition that can be motivated on purely classical grounds. Any finite spatial region bounded by a surface of finite area admits a finite number of independent field modes that can couple through that surface. The discreteness scale $\epsilon$ parameterizes this finite resolution: it is the minimal cell size below which no independent degree of freedom can be dynamically coupled through the partition boundary. In the cosmological application, $\epsilon$ is not a free parameter but is fixed by self-consistency (§6): requiring the emergent quantum description to be thermodynamically consistent with the classical horizon that produced it yields $\epsilon = 2\,l_p$ exactly.

The result $\epsilon = 2\,l_p$ is a *consequence* of the framework, not an input. The Planck length itself is a combination of $c$, $G$, and $\hbar$ — but here $\hbar$ is derived (§5.2), so $l_p$ is determined by the same self-consistency loop. The apparent circularity is broken because the classical ingredients ($c$, $G$, and $\epsilon$) enter before $\hbar$ is defined, and the relationship $\epsilon = 2\,l_p$ emerges after $\hbar$ has been fixed. No quantum concept is presupposed.

The deeper question is whether any framework claiming to derive quantum mechanics must posit *some* form of discreteness to avoid an infinite-dimensional configuration space — in which case the finiteness condition would be a necessary structural ingredient rather than an ad hoc addition. Holographic entropy bounds [27] provide independent, non-quantum motivation: the maximum entropy of a region bounded by area $A$ is $A/(4\,l_p^2)$, suggesting that finite-area boundaries enforce finite-dimensionality regardless of the theory used to describe the interior.

## 9.5 Observable Vacuum Energy Effects

If the quantum vacuum energy is "compression noise" — a feature of the emergent description rather than a source term in Einstein's equations — a natural objection arises: why do vacuum energy effects such as the Casimir force and the Lamb shift produce observable, measurable consequences?

The answer is that these effects are predictions *within* the emergent quantum description, and that description is empirically exact for all visible-sector physics. The Casimir force arises from differences in zero-point energy between boundary configurations — it depends on energy *differences* between geometries, not on the absolute value of the vacuum energy density. The Lamb shift depends on virtual photon fluctuations coupling to atomic energy levels — again, a relative effect within the quantum description. Neither requires that the absolute zero-point energy be a gravitational source.

The distinction is between the *internal consistency* of the emergent description and its *gravitational coupling*. The emergent quantum mechanics governs all visible-sector dynamics: every calculation within it — scattering amplitudes, energy level shifts, force laws — is valid because the P-indivisible stochastic process faithfully reproduces unitary quantum mechanics (§3.1). The Casimir effect and the Lamb shift are among these internal predictions. What the framework denies is that the absolute vacuum energy computed within this description should appear as a source term in Einstein's equations, because the metric is part of the classical substratum that is logically prior to the quantum description (§7.2). Gravity sees the classical stress-energy, not the emergent zero-point sum. The $10^{122}$ discrepancy reflects the compression ratio, not a missing cancellation.

## 9.6 Relation to Prior Work

The framework shares motivations with several existing programs while differing in structure and conclusions.

**'t Hooft's cellular automaton interpretation [1]** is the closest predecessor. 't Hooft proposes that quantum mechanics emerges from deterministic dynamics through information loss at the Planck scale, with specific models based on cellular automata evolving under discrete, local rules. The present framework agrees that quantum mechanics can emerge from underlying determinism, but differs in three respects. First, the mechanism: 't Hooft's information loss is a feature of the microscopic dynamics, while the present framework's trace-out is a feature of the observer's epistemic situation — different observers with different partitions would see different quantum descriptions (§1.4), a feature absent from the cellular automaton program. Second, the generality: the present theorem applies to any system satisfying the axioms and conditions, independent of specific microphysical content, while 't Hooft's models require particular dynamical rules. Third, the relationship to Bell's theorem: 't Hooft's framework requires superdeterminism (violation of measurement independence), while the present framework preserves measurement independence and violates outcome independence through P-indivisible joint dynamics (§3.2).

**QBism and relational quantum mechanics** treat quantum states as expressions of an agent's beliefs (QBism) or as relative to interacting subsystems (RQM). The present

framework agrees that the quantum state is observer-relative and epistemic in character, but provides a structural account of *why*: the quantum description arises from a specific mathematical operation (the trace-out) applied to a specific physical situation (the causal partition), and its observer-dependence is a derived consequence (partition-relativity) rather than an interpretive starting point. The framework also makes quantitative predictions — the value of $\hbar$, the dissolution of the cosmological constant problem, dark energy evolution — that purely interpretive programs do not.

**Jacobson's thermodynamic derivation of Einstein's equations [16]** showed that the Einstein field equations follow from thermodynamic reasoning applied to local Rindler horizons, treating spacetime geometry as an equation of state. The present framework inverts the direction: where Jacobson derives gravity from thermodynamics, the present framework derives quantum mechanics from the causal structure that gravity creates. The two results are complementary — each takes one of the two theories as input and derives constraints on the other from the observer's epistemic situation — and their consistency is a non-trivial check on both.

**AdS/CFT and holographic programs.** The framework offers a structural explanation for the success of the AdS/CFT correspondence. In AdS/CFT, the boundary CFT provides a complete description of the bulk — the boundary observer is *external* to the system it describes, immune to the inferential limits that Wolpert [19] proved constrain embedded observers. The holographic description works precisely because the boundary observer's causal access is not restricted by a partition of the kind the theorem requires. This is consistent with, but independent of, the standard derivations of AdS/CFT from string theory.

## 9.7 Assumptions, Limitations, and Falsifiability

**Assumptions and limitations of the theorem.** The finiteness condition is an independent assumption in Part I. In Part II, the cosmological horizon's finite area provides finiteness through the partition geometry, so it is not an additional assumption in the physical application. The theorem is silent on systems where neither an intrinsic discreteness scale nor a finite-area partition boundary ensures finite-dimensionality. Barandes' stochastic-quantum correspondence is the mathematical engine of the theorem, entering at a single step (§3.1). Its proof rests on the Stinespring dilation theorem and has been independently corroborated (§9.3).

**Assumptions and limitations of the application.** The derivation of $\hbar = c^3\epsilon^2/(4G)$ (§5.2, Steps 1–4) rests on four components, each placed on rigorous footing: phase uniqueness (D-gauge completeness theorem, §5.3), volume-independence (volume-independence lemma, §5.2), scale determination via thermodynamic equilibrium (which works within finite-dimensional quantum statistical mechanics, requiring no continuum QFT), and lat-

tice robustness (corrections bounded by $\mathcal{O}(10^{-122})$). The remaining genericity assumption — non-degenerate spectrum and non-vanishing configuration-basis overlaps — holds for all but a measure-zero set of systems satisfying (C1)–(C3). The tensor product structure of the emergent QFT (§7.1) follows from spatial locality of the dynamics and is standard for lattice Hamiltonian systems. A rigorous derivation showing that the Barandes correspondence preserves spatial locality is expected to hold generically but is not proved here. This step is required for the CC dissolution (§7) but not for the derivation of $\hbar$ (§5). The echo prediction relies on a kinematic argument; the exact reflection amplitude requires non-perturbative calculation. The coefficient $\nu_{\mathrm{OI}}$ is constrained to be positive and small ($\nu_{\mathrm{OI}} \ll 1$) by the framework's internal logic (§8.1), consistent with phenomenological fits at $\mathcal{O}(10^{-3})$. Its exact value requires non-perturbative methods.

**The Planck-scale ordering question.** The claim that spacetime geometry is logically prior to the quantum description (§7.2) is well-motivated within the framework: the causal partition is defined by null geodesics, so the metric must exist before the partition that produces quantum mechanics. However, this ordering may break down at the Planck scale, where the distinction between "classical geometry" and "quantum description" could lose operational meaning. The framework does not address the deep Planck regime — its domain of validity is the macroscopic causal structure of the cosmological horizon — and a satisfactory account of trans-Planckian physics remains an open question. The framework's internal consistency does not require resolving this question: the self-consistency loop (§6) fixes $\epsilon$ at $2\,l_p$, above the scale where the ordering might break down, and the predictions (§8) probe regimes well within the framework's domain.

**What would falsify each part independently.** The *theorem* would be falsified by a restriction of Barandes' correspondence that excludes the class of processes generated here, or by a demonstration that the finiteness condition cannot be satisfied in the relevant physical regime. The P-indivisibility result (§2.3) is a mathematical theorem for finite systems and is not subject to empirical falsification; its physical relevance depends on the finiteness condition and the validity of conditions (C1)–(C3) in the system under consideration. The *application* would be falsified by the definitive absence of dark energy evolution in precision cosmological surveys, or by the confirmed absence of gravitational wave echoes at the predicted timescale, while leaving the theorem intact.

# 10 CONCLUSION

Two results at different levels of generality have been established.

**The general theorem (Part I).** Under three axioms (deterministic dynamics, a causal partition, and classical probability) and a finiteness condition on the effective configuration space, the P-indivisibility of the reduced dynamics is proved as a math-

ematical theorem: any bijective dynamics on a finite system with non-trivial coupling (condition C1) necessarily produces P-indivisible stochastic dynamics in the visible sector. Conditions (C2) and (C3) — inverted timescale separation and sufficient capacity — ensure the physical persistence and observational relevance of this indivisibility. By Barandes' stochastic-quantum correspondence, the embedded observer's reduced description is necessarily quantum mechanics. Crucially, the continuous-time transition data rigorously locks the quantum phases, resolving the Schur–Hadamard gauge freedom and leaving no physical ambiguity in the emergent unitary evolution. The Schrödinger equation, the Born rule, and Bell inequality violations are structural consequences. The action scale $\hbar$ is uniquely determined by the local structural properties of the partition boundary. No quantum postulates are required. The characterization theorem (§3.3) establishes that the conditions are not merely sufficient but *necessary*: any P-indivisible process, realized through a deterministic dilation, requires non-trivial coupling (C1), slow-bath memory (C2), and sufficient hidden-sector capacity (C3). Quantum mechanics and embedded observation under (C1)–(C3) are equivalent.

**The cosmological application (Part II).** The cosmological horizon provides a physical realization where the sharp-partition approximation is exact and all three conditions are satisfied. The theorem yields: (a) the exact value $\hbar = c^3\epsilon^2/(4G)$ from partition-relativity, the structural/volumetric distinction, dimensional analysis, and thermal self-consistency, with $\epsilon = 2\,l_p$ and the Bekenstein–Hawking entropy formula $S = A/(4\,l_p^2)$ recovered as consequences; (b) a fundamental dissolution of the cosmological constant problem, recognizing the $10^{122}$ discrepancy not as a catastrophic fine-tuning failure, but precisely as $S_{\mathrm{dS}}$ — the information compression ratio of the observer's quantum description; and (c) falsifiable predictions including dark energy evolution in RVM form and gravitational wave echoes.

The universe is not broken. The apparent conflict between its two most successful theories is the information-theoretic cost of observing it from within.

# DECLARATION OF AI-ASSISTED TECHNOLOGIES

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